Tópicos Especiais - Introdução À Manufatura Aditiva Para Metais

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Aluno especial
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Eng. MsC. Geociências

Modelagem matemática

- Modelos de fluxo de águas subterrâneas em meios porosos - MODFLOW
- Fluodinâmica Computacional - CFD

Eng. Mecânica

Título do Projeto

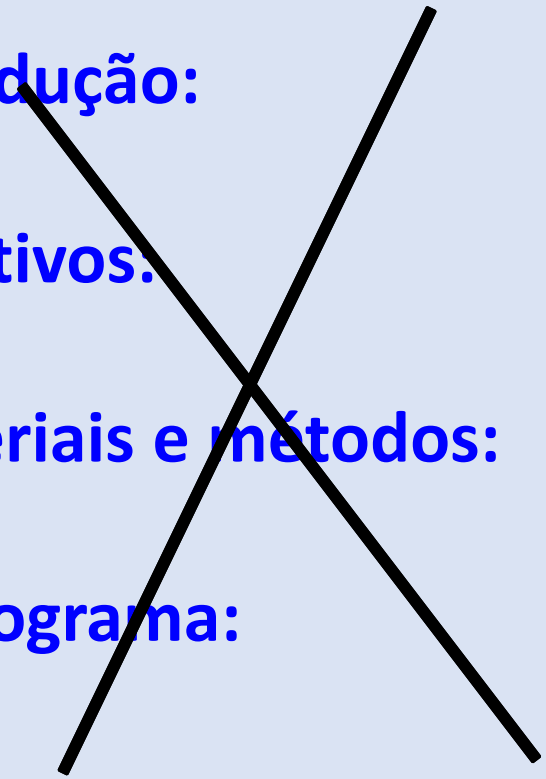
Introdução:

Objetivos:

Materiais e métodos:

Cronograma:

Título do Projeto



Introdução:

Objetivos:

Materiais e métodos:

Cronograma:

O que é um modelo?

Para que serve um modelo?

O que modelar?

Como fazer um modelo... ?

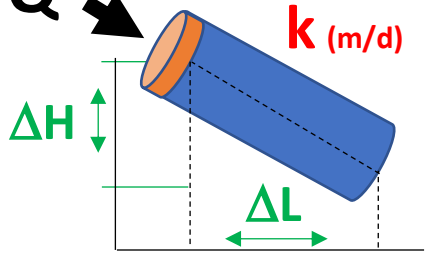
Onde... ?

Qual software usar, e porquê ?

Contexto

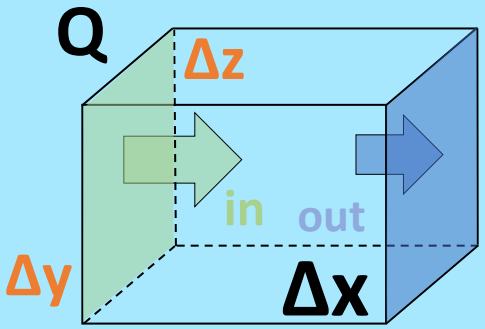
H_2O Calor

Darcy ~ Fourier



$$\frac{Q}{A_{\varnothing}} = q = k\frac{\Delta H}{\Delta L}$$

$$\dot{Q} = (-kA\,\Delta T)/L$$



$$\frac{\partial q_x}{\partial x}\,\Delta x\,\Delta y\Delta z$$

$\Delta x\ \Delta y\ \Delta z$ …

$\Delta x\ \Delta y\ \Delta z$

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} + \frac{\partial^2 h}{\partial z^2} = Q + PA$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial h}{\partial x}\right)$$

Laplace

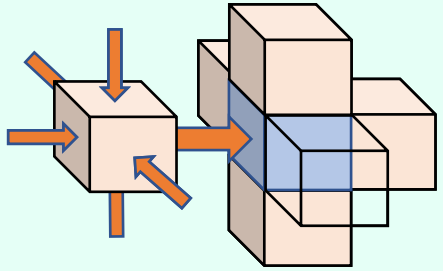
$$\nabla^2 h$$

nabla

= fontes

(-) sumidouros

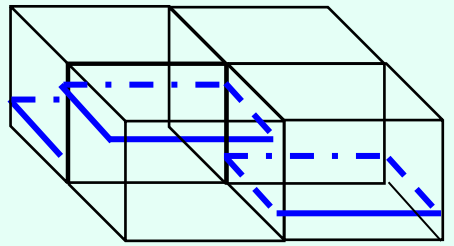
MODFLOW



O que é um modelo?

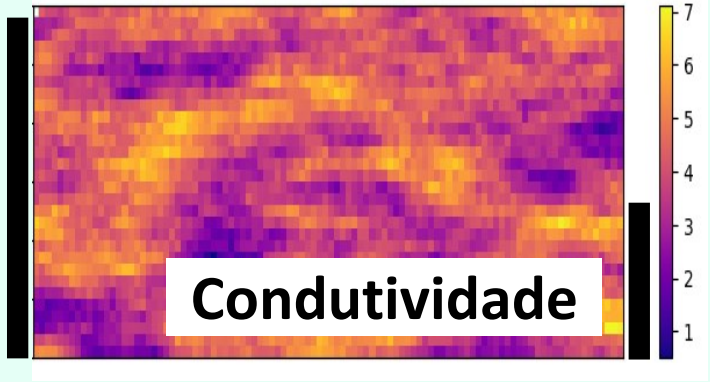
numérico

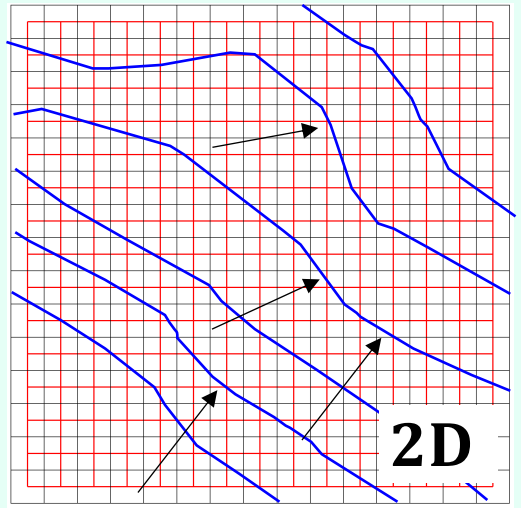
infinitesimal



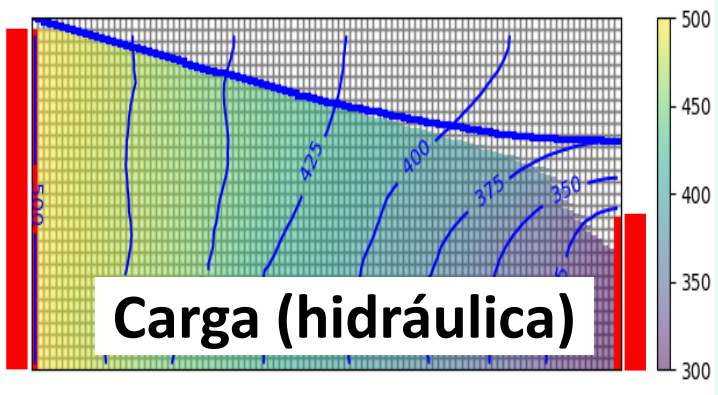
H_n

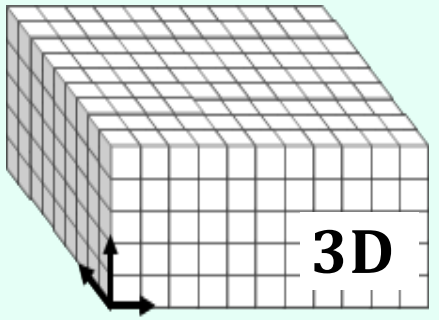
Premissas

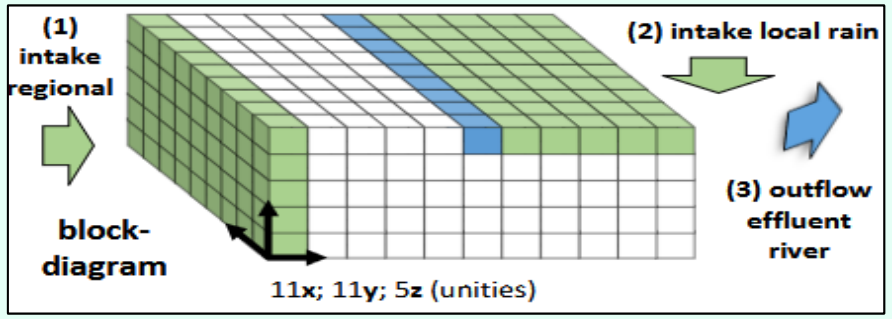




Resultados







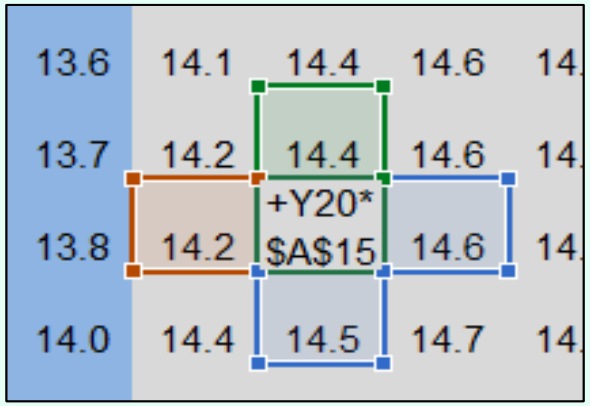
2D

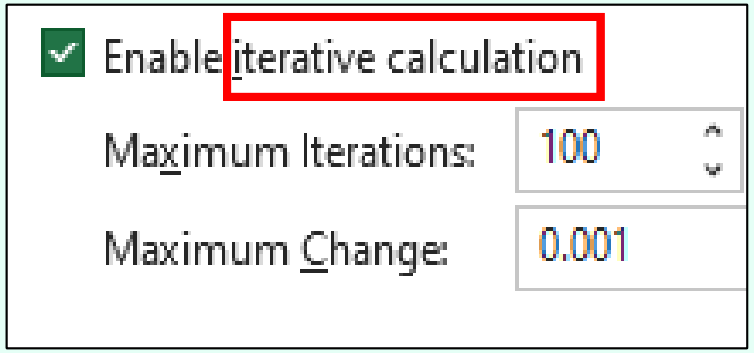
$$A = \frac{B + C + D + E}{4}$$

Método de Diferenças finitas (FDM)

Table 2: Cross-section of the 10th y-row in table form

	1	2	3	4	5		1	2	3	4	5
1	14.8	14.8	14.7	14.6	14.6	14.6	14.7	14.9	14.9	15.0	15.0
2	14.8	14.8	14.7	14.6	14.6	14.6	14.7	14.8	14.8	14.9	14.9
3	14.8	14.8	14.7	14.6	14.6	14.6	14.7	14.7	14.8	14.8	14.8
4	14.8	14.8	14.7	14.6	14.6	14.6	14.7	14.7	14.7	14.8	14.8
5	14.8	14.8	14.7	14.6	14.6	14.6	14.7	14.7	14.7	14.7	14.8





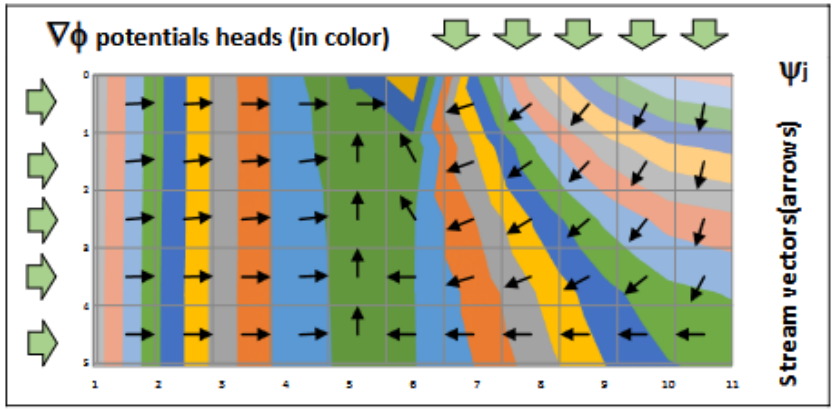
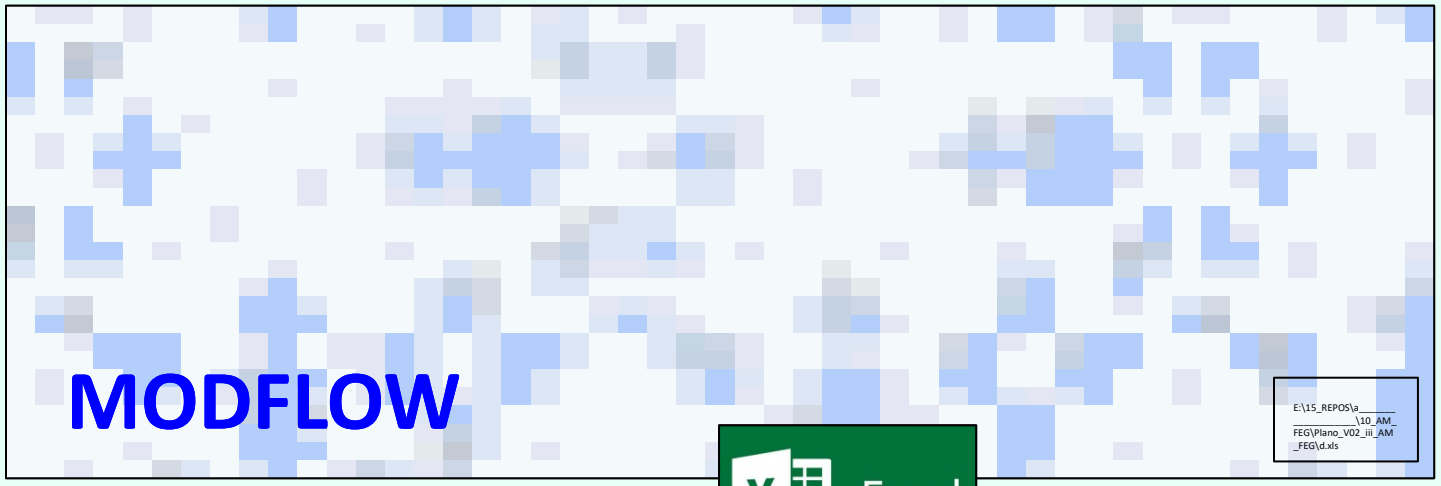


Fig 5: Cross-section of the 10th y-row in graphic form



convecção difusão fontes

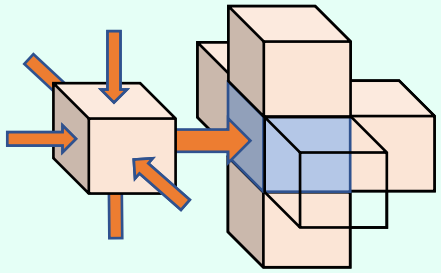
$$\frac{\partial(\rho\phi)}{\partial t}+\nabla.(u\phi)-[\nabla.(\Gamma_i\nabla)]^n\phi-S_\phi=0$$

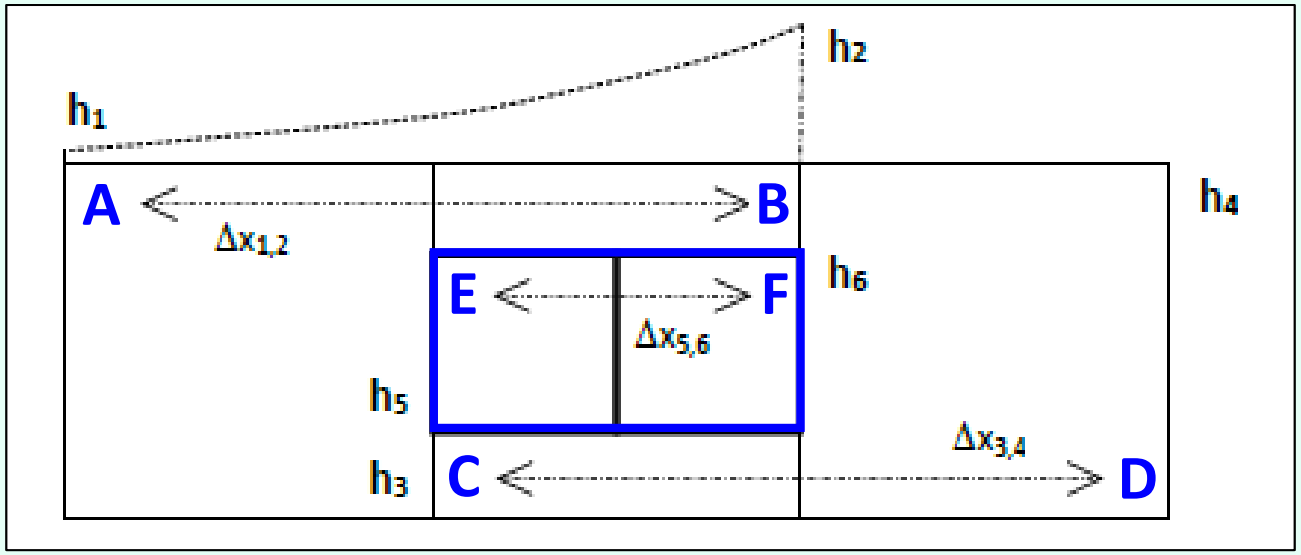
Método de Diferenças finitas (FDM)

$$\frac{\partial^2 h}{\partial x^2}+\frac{\partial^2 h}{\partial y^2}+\frac{\partial^2 h}{\partial z^2}=Q+PA$$

$$\nabla^2 h$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial h}{\partial x}\right)$$





$$\frac{\partial^2 h}{\partial x^2}(EF)\cong\frac{[(\Delta h_{1,2})/\Delta x_{1,2}]-[(\Delta h_{3,4})/\Delta x_{3,4}]}{\Delta x_{5,6}}$$

$$h_{i,j}=\frac{(h_N+h_S+h_E+h_W+h_T+h_B)}{6}-\frac{Q+PA}{6T}$$

https://drive.google.com/file/d/1SBsgUjAlBehVgu04VcFnFxR8JmSZ7hi6/view

PDF

" pausa para um comentário aqui ... "

CFD

Otimização Topográfica

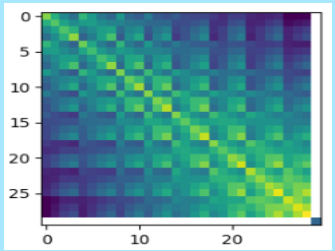
$$\frac{\partial \phi}{\partial t} + \nabla.(u\phi) - D\nabla^2\phi = S_\phi$$

convecção difusão fontes

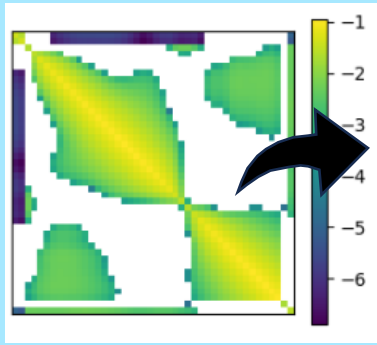
$$\begin{bmatrix} * & * & & * \\ * & * & * & \\ & * & * & * \\ * & & * & * \end{bmatrix} * \begin{bmatrix} \phi \end{bmatrix} = \begin{bmatrix} * \\ * \\ * \\ * \end{bmatrix}$$

$$A * \phi = b$$

Modelo OBS ⟷ PAR



COV



0.4	0.1	0.2
0.1	0.4	0.2
0.2	0.2	0.4

Zoom +

Modflow

PEST!

Notações:

Diferencial x Matricial

Condutividade hidráulica

x load flow

- Densidade (Massa)
- Elasticidade
- Rigidez
- Limite de escoamento

Probabilidade condicional

$$\phi = A^{-1} b$$

Inversão

Sparce matrix

Derivative matrix

Asymmetric matrix

COVariance matrix

Coefficient matrix

Regressor matrix

Design matrix

Jacobian matrix

Abordagem Estocástica

5.2 Gauss-Seidel method

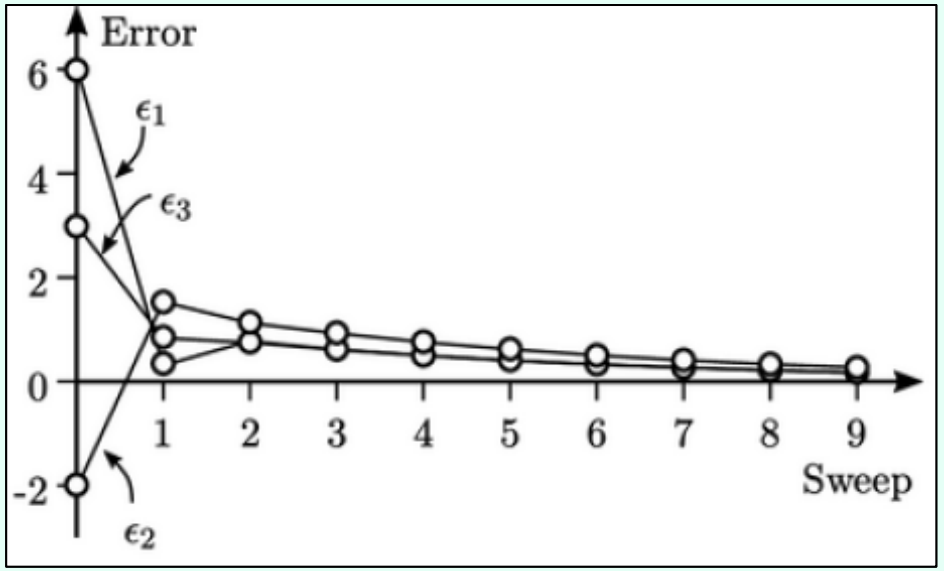
$$\mathbf{A} \quad * \quad \phi \quad = \quad \mathbf{b}$$

$$\begin{bmatrix} 3 & -1 & -1 \\ -1 & 5 & -2 \\ -1 & -2 & 4 \end{bmatrix} \begin{bmatrix} \Psi_1 \\ \Psi_2 \\ \Psi_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix}.$$

$$\Psi_1 := \frac{1}{3}(6 + \Psi_2 + \Psi_3);$$
$$\Psi_2 := \frac{1}{5}(-2 + \Psi_1 + 2\Psi_3);$$
$$\Psi_3 := \frac{1}{4}(3 + \Psi_1 + 2\Psi_2).$$

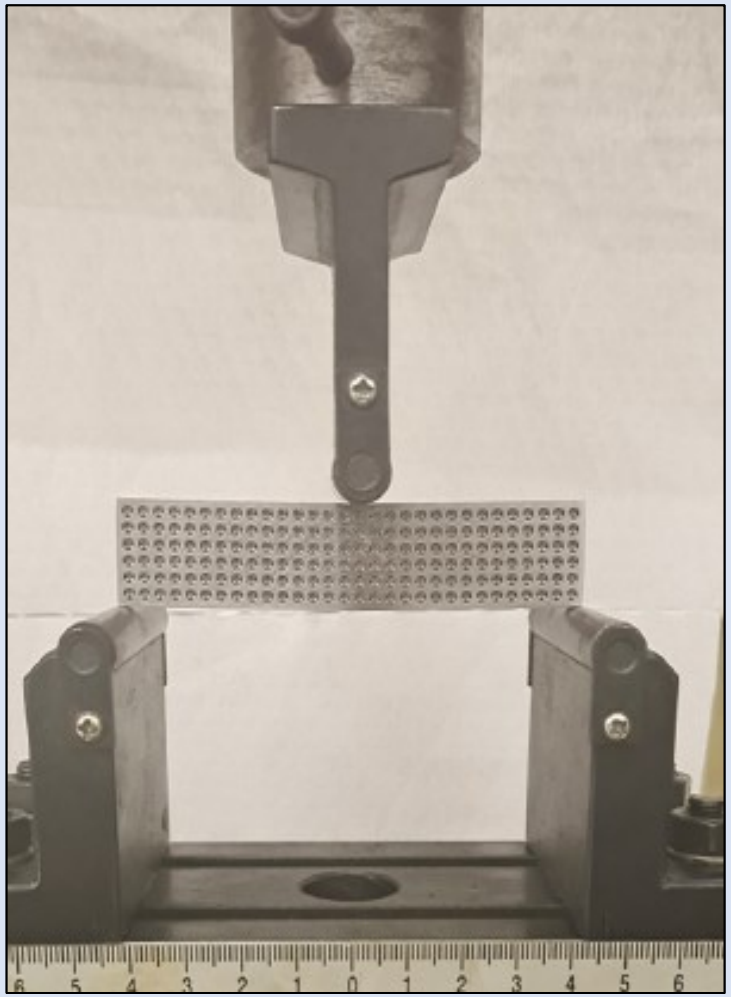
Variable	Start	Sweep 1	...2	...3	...4
Ψ_1	0.0000	2.0000	2.4167	2.7431	2.8821
Ψ_2	0.0000	0.0000	0.5833	0.8069	0.9121
Ψ_3	0.0000	1.2500	1.6458	1.8392	1.9266

$$\Psi_i := \frac{1}{a_{i,i}} \left(b_i - \sum_{\substack{j=1 \\ i \neq j}}^{N} a_{i,j} \Psi_j \right)$$



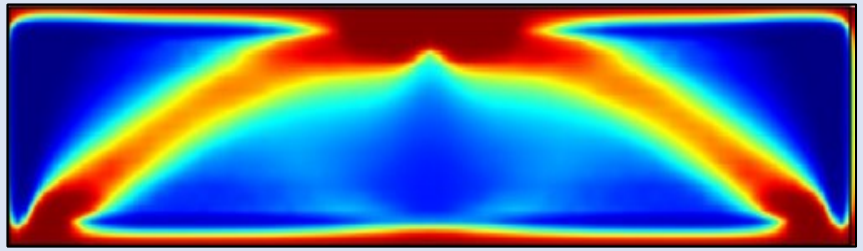
“ pausa para um comentário aqui ... ”

Experimento:

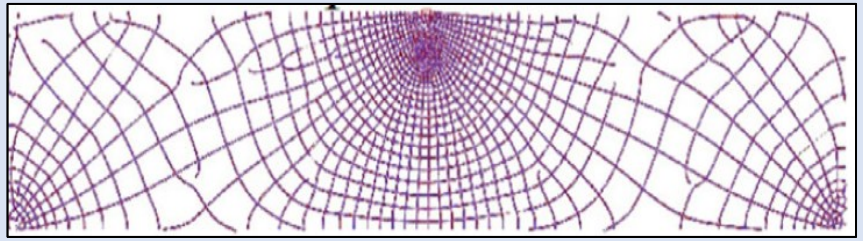


Modelo matemático

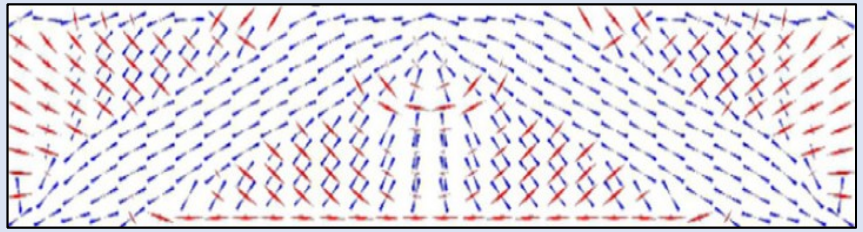
Otimização Topográfica



Fluxo ... de tensões



Mapa potenciométrico



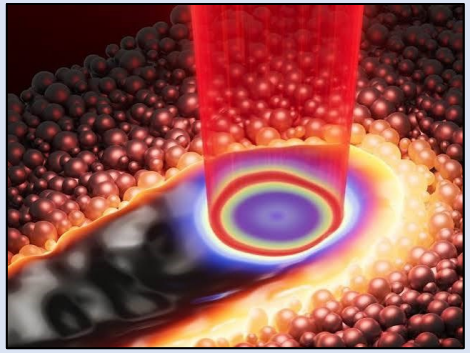
Vetores



Peça final

Metamaterials mapped lightweight structures by principal stress lines and topology optimization: Methodology, additive manufacturing, ductile failure and tests - Liu et. al (2021)

Fusão a laser em leito de pó







Manufatura Aditiva Para Metais

Arco Voltaico

Soldas MIG - TIG

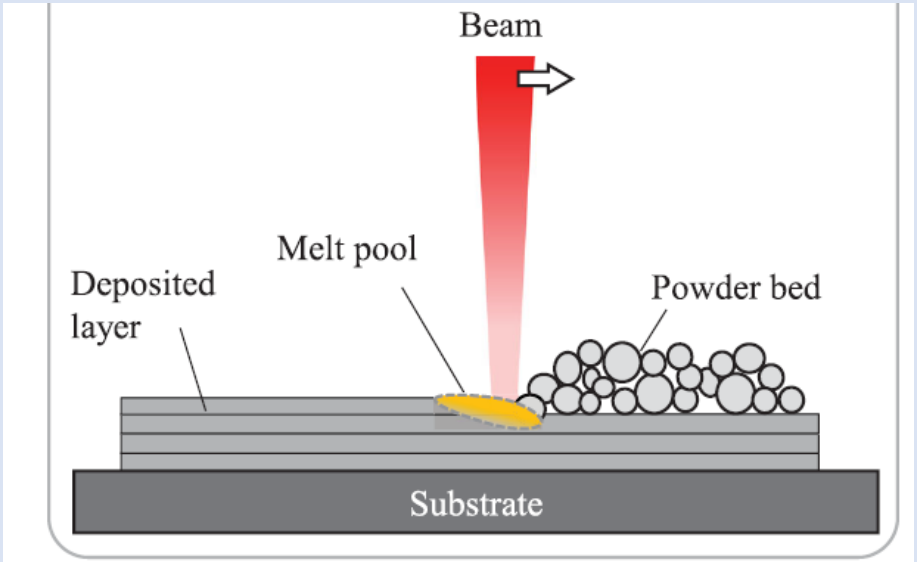


Metal
Inert
Gas
… el. tungsten

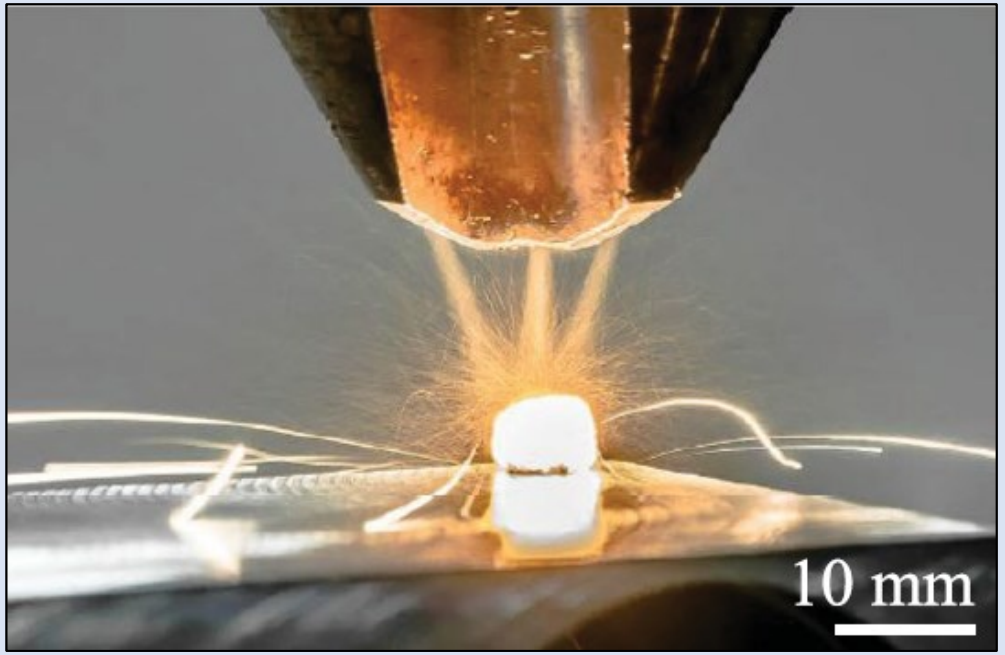


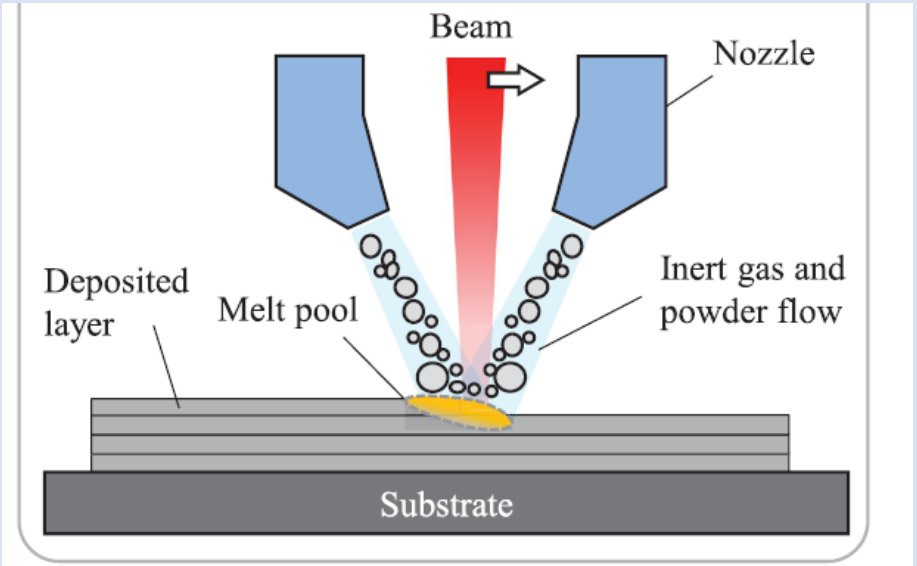
Powder Bed Fusion (PBF)





Directed Energy Deposition (DED)





Powder Bed Fusion (PBF)

Electron Beam Melting	EBM
Vat Photopolymerization	VPP
Binder Jetting (glue)	BJ
Material Jetting	MJ
Sheet Lamination (cut)	SHL

Classificações A.M.:

- Por fonte de energia
- Por tipo de material

Directed Energy Deposition (DED - ARC)



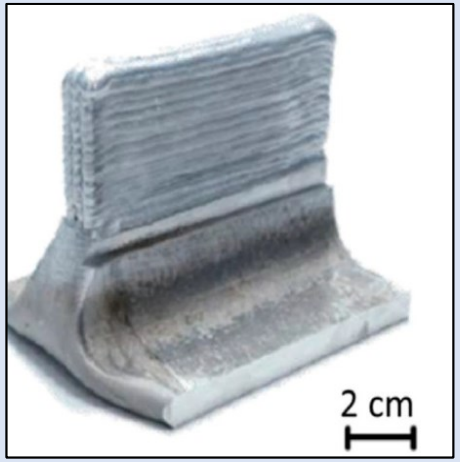
GTAW

Wire Arc

WAAM

Wire DED

Arco Voltaico



Modelling and simulation of metal additive manufacturing processes with particle methods: A review https://doi.org/10.1080/17452759.2023.2274494

PROBLEMAS A.M.

DED - Arco Voltaico

1- Orientação – intensidade ..da solda

2- Descontínuidades (*balling*)

3- Keyholes (espaços vazios - vapor)

4- Degradação química e oxidação

5- Partículas residuais

6- Trincas

7- Delaminação / Distorção

8- Acabamento superficial

9- Porosidade

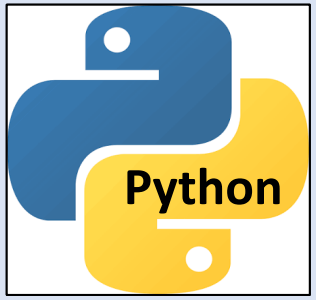
Para que modelar?

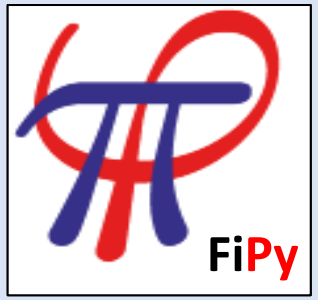
- **Racionalizar o problema;**
- **Criar hipóteses**
- **... entender a base de dados;**
- **Controlar o processo de fabricação.**

Grande quantidade de variáveis !

```
from fipy import Grid1D
         import numpy as np
         import matplotlib.pyplot as plt
```

finite volume method (FVM)





Linguagem orientada a objetos:

```
C1  nx=50 ;dx=1.                ; mesh=Grid1D(nx=nx,dx=dx)
C2  plt.figure(figsize=(20, 0.5))
C3  FF = mesh.faceCenters()     ; X=mesh.faceCenters[0]
C4  plt.xlim(FF.min(),FF.max()) ; plt.xticks(X)
C5  plt.plot  (mesh.cellCenters()  , np.zeros_like(mesh.cellCenters())
    ,'o-'                          , markersize=5      ,color='green')
C6  plt.grid(True, axis='x'        , linestyle='--'    ,color='red')
```

Visual Studio

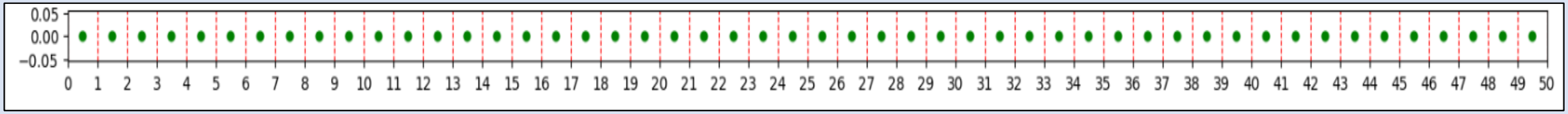


Jupyter Notebook





Resultado: GRID 1D



E:\02_REPOS\a_____\09_CFD_ON\02_FiPy

https://github.com/grossiRM/CFD_ON/blob/main/02_FiPy/01_mesh1D_Beta_Diffusion.ipynb

convecção difusão fontes

$$\frac{\partial(\rho\phi)}{\partial t} + \nabla.(u\phi) - [\nabla.(\Gamma_i \nabla)]^n \phi - S_\phi = 0$$

$$\frac{\partial \phi}{\partial t} = D\nabla^2\phi$$

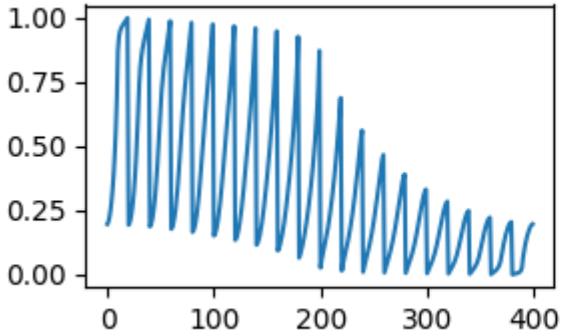
Tempo transiente difusão

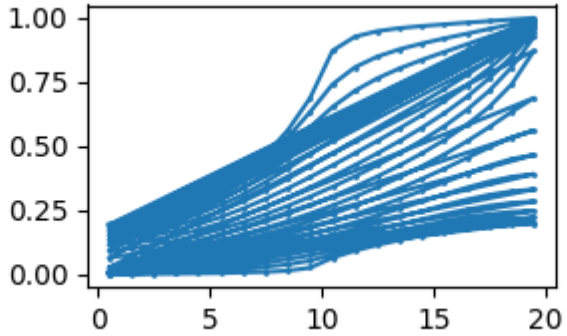
```
mesh=Grid2D (dx=1,dy=1,nx=20,ny=20)
```

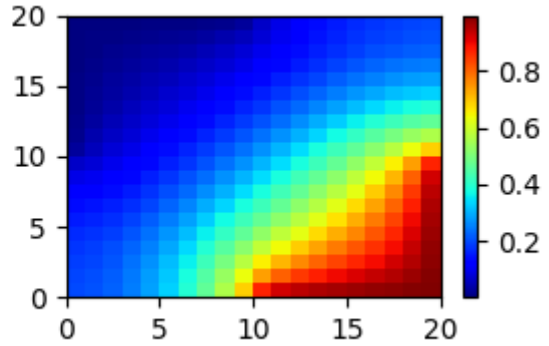
```
timeStepDuration = 10 * 0.9 * dx**2 / (2 * D)        ; steps = 10
for step in range(steps):    eq.solve(var=phi ,dt=timeStepDuration)
```

```
YY = phi.value  ;XX=np.linspace(0,len(YY)-1,num=len(YY))  ; XA = mesh.cellCenters[0]
xx = mesh.cellCenters[0].value.reshape(nx, ny)            ; zz = phi.value
yy = mesh.cellCenters[1].value.reshape(nx, ny)            ; ZZ = zz.reshape(nx, ny)

fig, (ax1, ax2, ax3) = plt.subplots(1, 3,figsize=(9, 2))
ax1.plot(XX,YY)
ax2.plot(XA.value,phi.value,'o-',markersize=1)  ; WW=ax3.pcolormesh(xx,yy, ZZ,cmap='jet')
cbar = plt.colorbar(WW)                         ; plt.tight_layout()          ; plt.show()
```







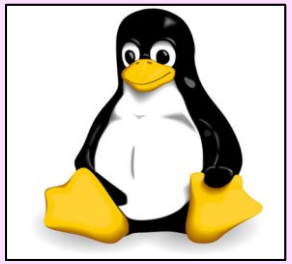
Resultado:

FiPy

https://github.com/grossiRM/CFD_ON/blob/main/02_FiPy/02_diffusion.mesh20x20.ipynb

PDEs

Partial Differential Equations



Linux



finite volume method (FVM)

```
==========
 \\      /   F ield
  \\    /    O peration
   \\  /     A nd
    \\/      M anipulation
```

Open

$$\nabla . \mathbf{V} = 0$$

Continuidade

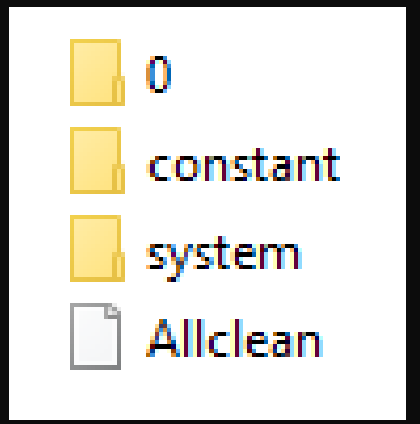
$$\rho \frac{\partial \mathbf{V}}{\partial t} = \rho \mathbf{g} - \nabla \mathbf{p} + \mu \nabla^2 \mathbf{V}$$

Qde de movimento

$$\rho C_V \frac{\partial T}{\partial t} = K \nabla^2 T + \Phi$$

Energia

```
.
├── 0
│   ├── T
│   ├── U
│   └── p
├── Allclean
├── constant
│   ├── momentumTransport
│   └── physicalProperties
└── system
    ├── blockMeshDict
    ├── controlDict
    ├── functions
    ├── fvSchemes
    └── fvSolution
```



C:\ D:\ E:\ ...

https://github.com/grossiRM/CFD_ON/blob/main/04_OpenFoam/03_NOTES-on-CFD_OpenFoam.pdf

PDEs

Partial Differential Equations

Método de Volumes finitos (FVM)

```
==========
 \\      /   F ield
  \\    /    O peration
   \\  /     A nd
    \\/      M anipulation
```

Open

Convection

source → 0

$$\frac{\partial \rho \Phi}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \Phi) = \nabla \cdot (D \nabla \Phi) + S_\Phi(\Phi)$$

Diffusion

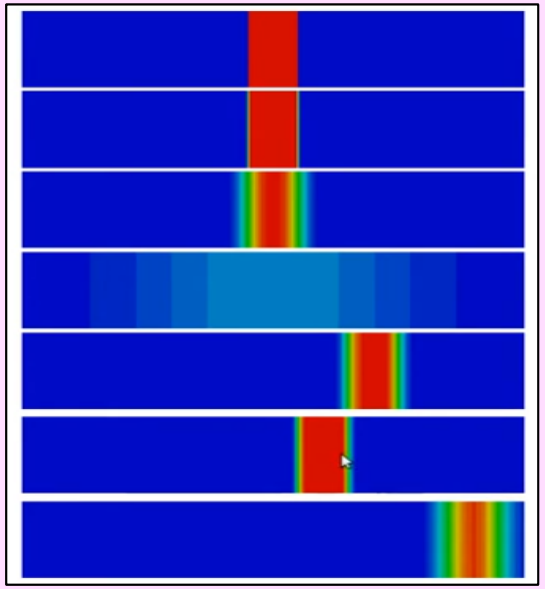
$$\frac{\partial \rho \Phi}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \Phi) = \nabla \cdot (D \nabla \Phi) + S_\Phi(\Phi)$$

0

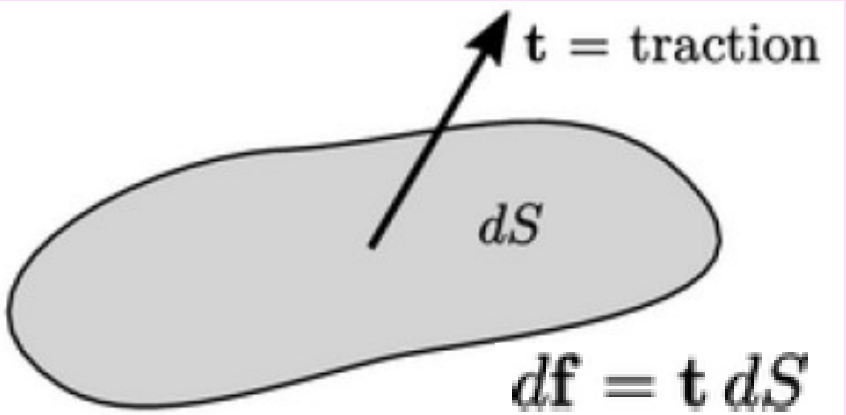
transient

Diffusion ⇔

Convection →



Método de Diferenças finitas (FDM)



$$\begin{pmatrix} \mathbf{t}_x \\ \mathbf{t}_y \\ \mathbf{t}_z \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix}$$

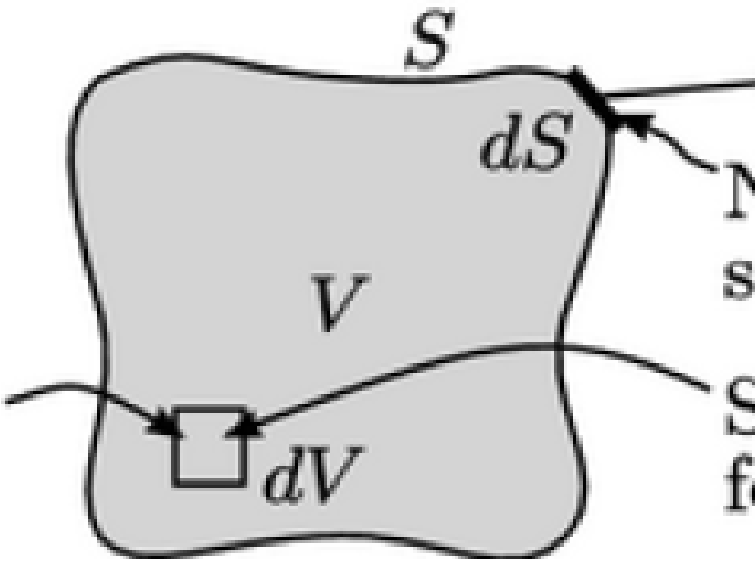
$$d\mathbf{f} = \mathbf{t}\,dS = (\mathbf{n}\bullet\boldsymbol{\sigma})\,dS = d\mathbf{S}\bullet\boldsymbol{\sigma}.$$

$$\int_S \mathbf{t}\,dS = \int_S \mathbf{n}\bullet\boldsymbol{\sigma}\,dS = \int_S (d\mathbf{S}\bullet\boldsymbol{\sigma})$$

$$\int_S (d\mathbf{S}\bullet\boldsymbol{\sigma}) = \int_V \nabla\bullet\boldsymbol{\sigma}\,dV$$

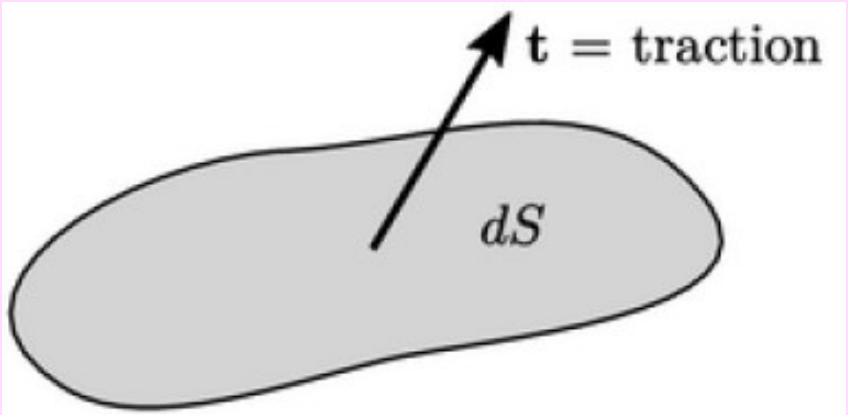
Rate of change of momentum

$$\frac{D}{Dt}\int_V \rho\mathbf{u}\,dV$$



Net force at surface $= \int_S \mathbf{t}\,dS$

Sum of body forces $= \int_V \rho\mathbf{b}\,dV$



$$\begin{pmatrix} \mathbf{t}_x \\ \mathbf{t}_y \\ \mathbf{t}_z \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix}$$

$$\boldsymbol{\sigma} = \begin{pmatrix} \mathbf{t}_x \\ \mathbf{t}_y \\ \mathbf{t}_z \end{pmatrix}$$

$$\nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\int_S (d\mathbf{S} \cdot \boldsymbol{\sigma}) = \int_V \nabla \cdot \boldsymbol{\sigma}\, dV$$

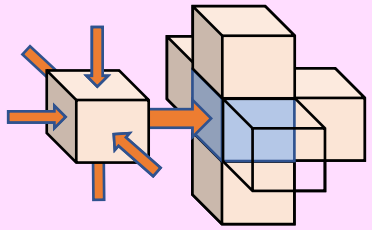
$$\int_V \left[\rho \frac{D\mathbf{u}}{Dt} - \nabla \cdot \boldsymbol{\sigma} - \rho \mathbf{b} \right] dV = 0.$$

Momento ...

Qde de movimento

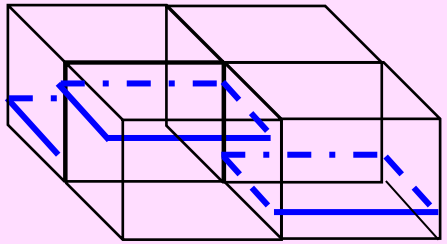
$$\nabla \cdot \boldsymbol{\sigma} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \int_S (d\mathbf{S} \cdot \boldsymbol{\sigma}).$$

Método de Diferenças finitas (FDM)

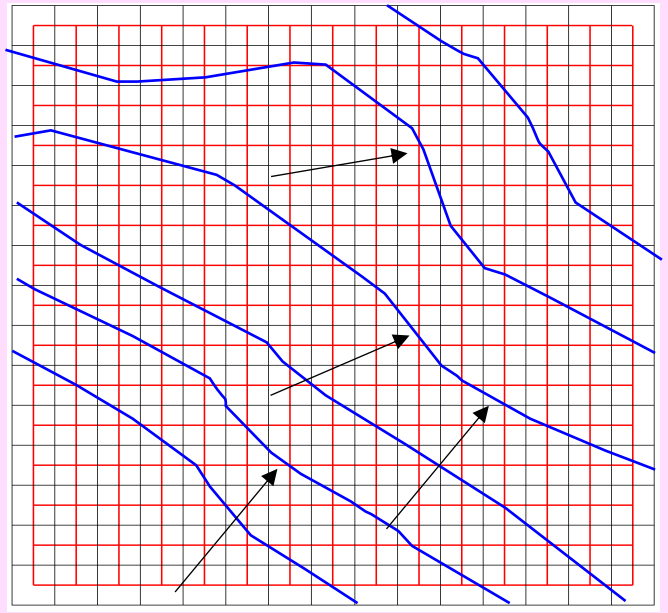


$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

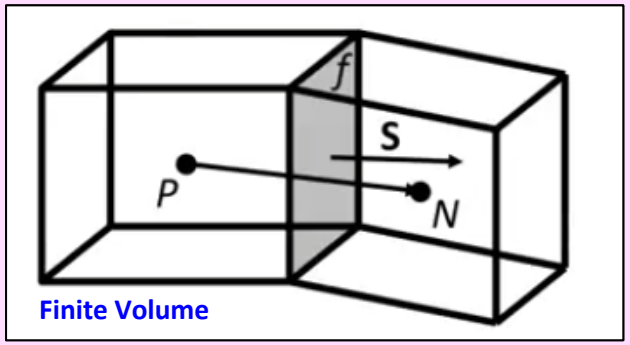
Séries de Taylor



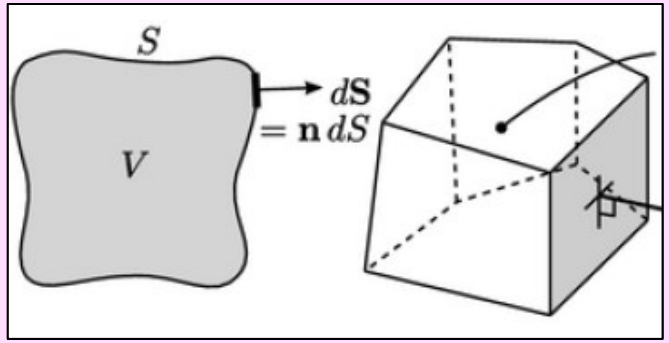
$\updownarrow H_n$



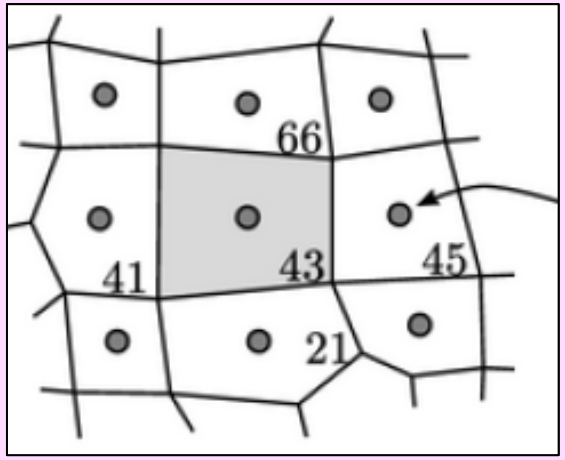
Método de Volumes finitos (FVM)



Finite Volume



$$\int_V \nabla \bullet (\rho \mathbf{U} \phi)\, dV = \int_S d\mathbf{S} \bullet (\rho \mathbf{U} \phi)$$



$$\approx \sum_f \mathbf{S} \bullet (\rho \mathbf{U})_f \phi_{f(F,S)}$$

$$= \sum_f F \phi_{f(F,S)}$$

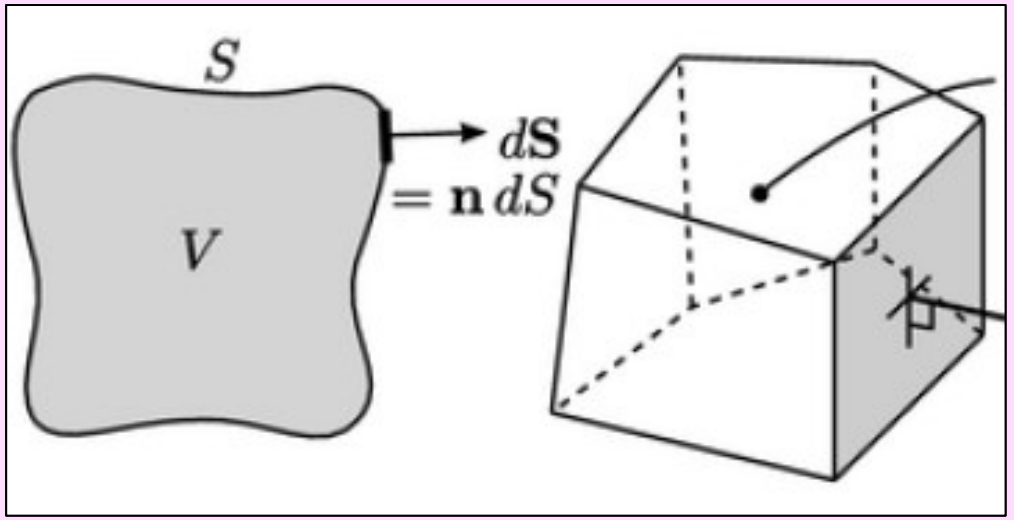
$$\frac{\partial(\rho\phi)}{\partial t} + \nabla.(u\phi) - [\nabla.(\Gamma_i \nabla)]^n \phi - S_\phi = 0$$

convecção difusão fontes

Volumes finitos (FVM)

Time Derivative	$\left[\left[\frac{\partial \rho[\phi]}{\partial t}\right]\right]$	$\frac{\rho_P^n \phi_P^n - \rho_P^o \phi_P^o}{\Delta t} V_P$
Convection Term	$\left[\left[\nabla \cdot \left(F[\phi]_{f(F,S)}\right)\right]\right]$	$\sum_f F \phi_{f(F,S)}^n$
Diffusion Term	$[[\nabla \cdot (\Gamma \nabla[\phi])]]$	$\sum_f \Gamma_f \mathbf{S} \cdot \nabla_f \phi^n$
Divergence Term	$\nabla \cdot \phi$	$\sum_f \mathbf{S} \cdot \phi_f^o$
Cell Gradient Term	$\nabla \phi$	$\sum_f \mathbf{S} \phi_f^o$
Curl	$\nabla \times \phi$	$2\,*(\text{skew}\,\nabla \phi^o)$
Implicit Source Term	$[[S_I[\phi]]]$	$S_I V_P \phi^n$
Explicit Source Term	S_E	$S_E V_P$
Cell Average	$\langle\phi\rangle_\square$	$\frac{1}{N}\sum_{P=\square} \phi_P^o$

Finite Volume Notation



Teorema de Gauss
Superfície ... Volume

$$\sum F \phi_{f(F,S)}$$

$$h_{i,j} = \frac{(h_N + h_S + h_E + h_W + h_T + h_B)}{6} - \frac{Q + PA}{6T}$$

PDEs

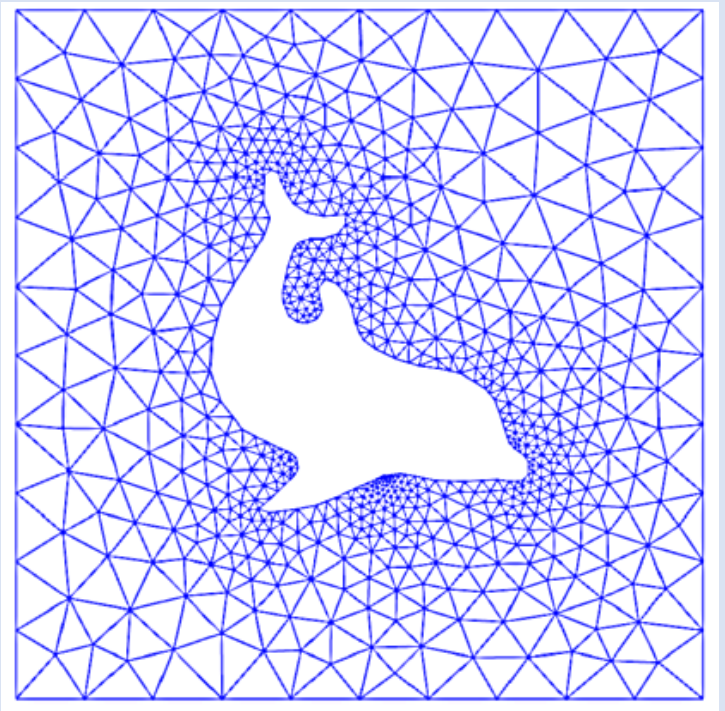
Partial Differential Equations



finite elements method (FEM)

$$u = \sum_{i=0}^{N} c_i \psi_i(x, y),$$

Malha triangular



$$\int_{\Omega_e} -\nabla^2 u \, v \, \mathrm{d}x = \int_{\Omega_e} f \, v \, \mathrm{d}x$$

$$\int_{\Omega_e} -\nabla^2 u \, v \, \mathrm{d}x = \int_{\Omega_e} \nabla u \cdot \nabla v \, \mathrm{d}x - \int_{\partial\Omega_e} \frac{\partial u}{\partial n} \, v \, dS$$

$$\int_{\Omega_e} \nabla u \cdot \nabla v \, \mathrm{d}x$$

Introduction to Numerical Methods for Variational Problems - Langtangen & Mardal

https://github.com/grossiRM/CFD_ON/blob/main/04_OpenFoam/FEniCS/01b__A%20FEniCS%20Tutorial__ii__.pdf

Heat Affected Zone (HAZ)

finite elements method (FEM)



multi-timestep method

the balance of energy equation

+ phase change

+ growing domain

born-dead element technique

Simulação do processo

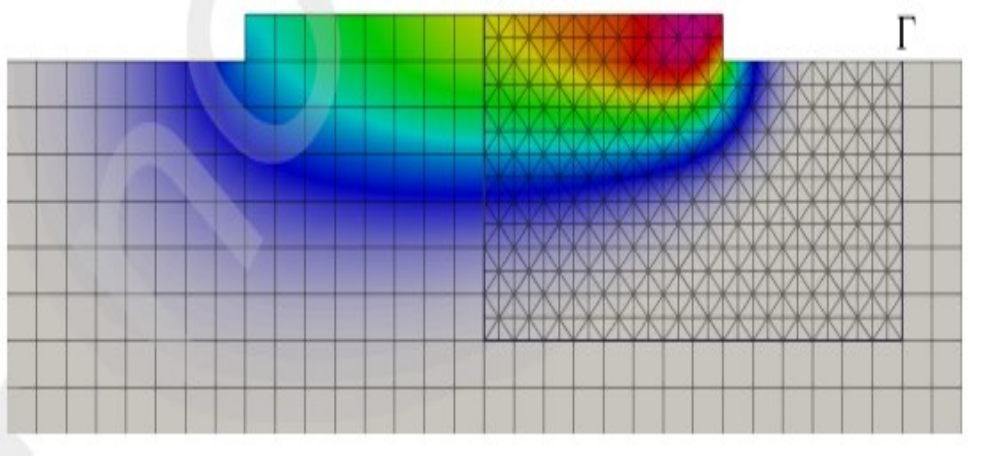
Digital twin

moving mesh

moving heat source

Adaptive Mesh Refinement





Exemplo:

https://github.com/ordinary-slim/moving_heat_source

https://github.com/ordinary-slim/mhs_fenicsx

A Chimera method for thermal part-scale metal Additive Manufacturing simulation

Substepped and advected **subdomain methods** for part-scale LPBF modeling

“ pausa para um segundo comentário aqui ... ”

Ecossistema

https://github.com/usnistgov

usnistgov

Overview | Repositories 1.4k | Projects 5 | Packages | People 40

National Institute of Standards and Technology

Department of Commerce

2k followers | Gaithersburg, Md. | https://www.nist.gov

FiPy

https://github.com/grossiRM/CFD_ON

CFD_ON

main | 1 Branch | 0 Tags

grossiRM 14

01_DeGroot	02
02_FiPy	07
03_OBJ_FUNC	02
04_OpenFoam	14
05_Linkedin_Posts	08
06_PyLops	04
_aux	04
.gitignore	10
README.md	06
desktop.ini	02
w_14_control_.txt	14

https://github.com/new

Create a new repository

Repositories contain a project's files and version history.

Required fields are marked with an asterisk (*).

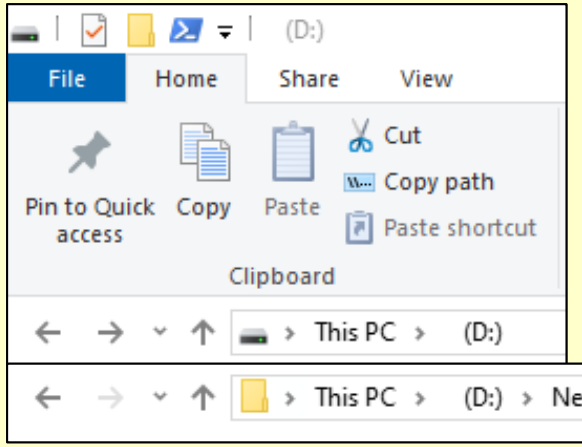
1 General

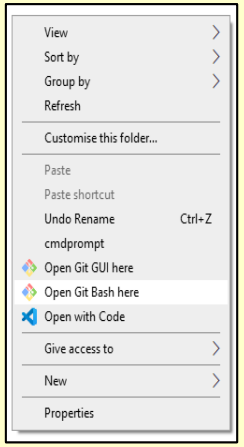
Owner * grossiRM / Repository name * new

new is available.

...or create a new repository on the command line

```
echo "# new" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/grossiRM/new.git
git push -u origin main
```

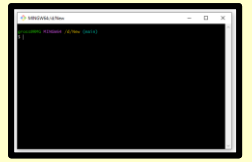


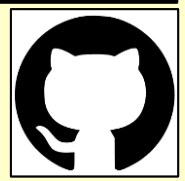


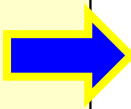

```
echo "# new" >> README.md
git initgit add README.md
git commit -m "first commit“
git branch -M main
git remote add origin https://github.com/grossiRM/new.git
git push -u origin main
```



```
$ git add .
$ git commit -m “001"
$ git push origin
```



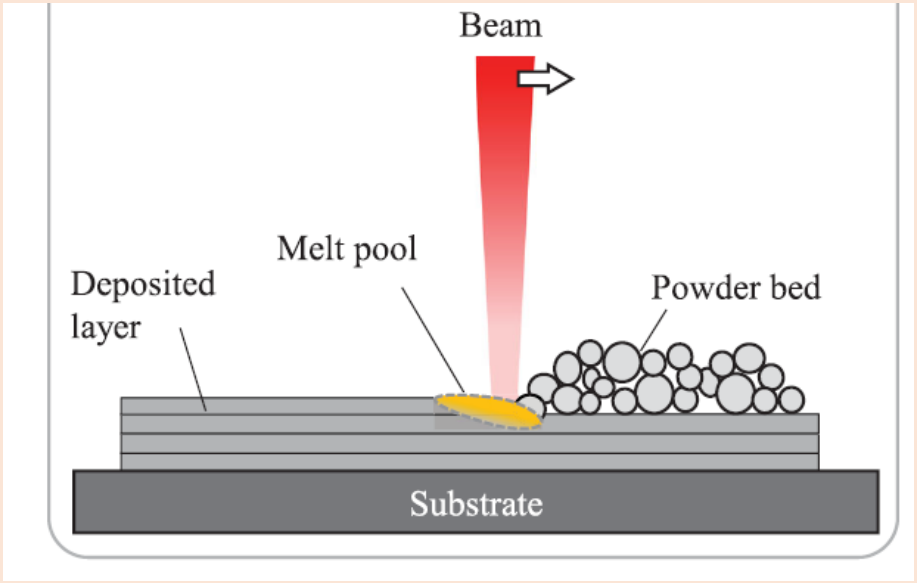




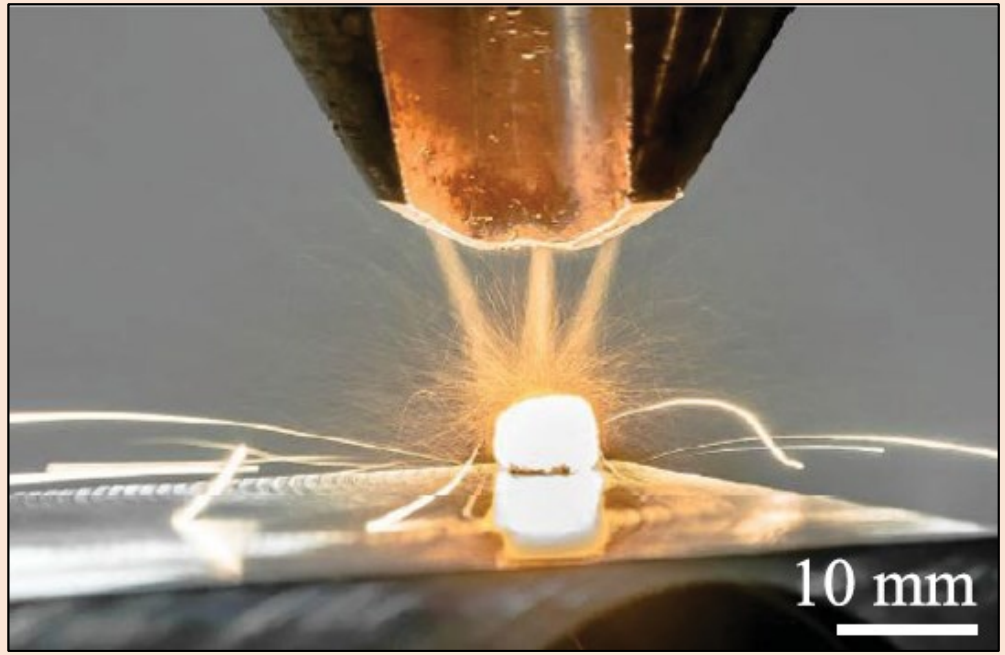
Name	Type
.git	File folder
README.md	Markdown Source File

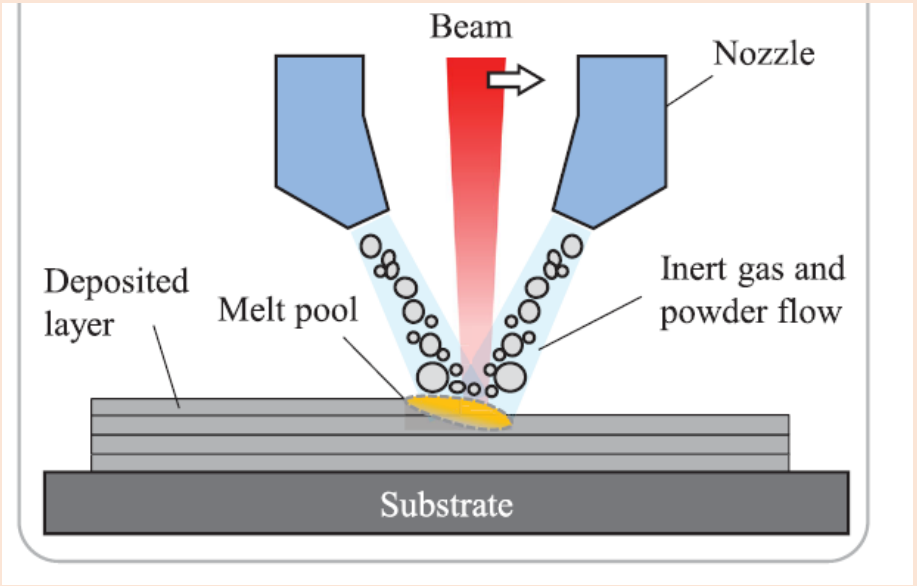
Powder Bed Fusion (PBF)





Directed Energy Deposition (DED)







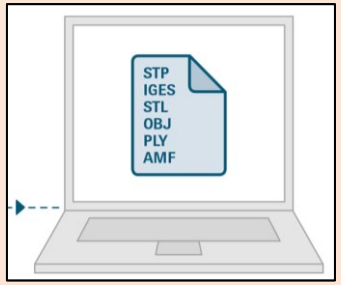
O que modelar?

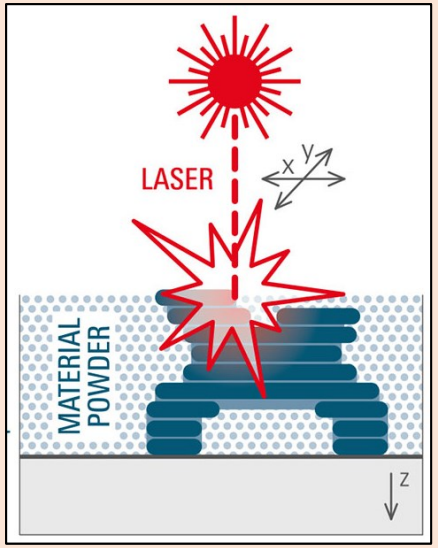
CAD (Computer-Aided Design)

shapes - parametric 3D models (length, width, angles + Rules + History Tree)



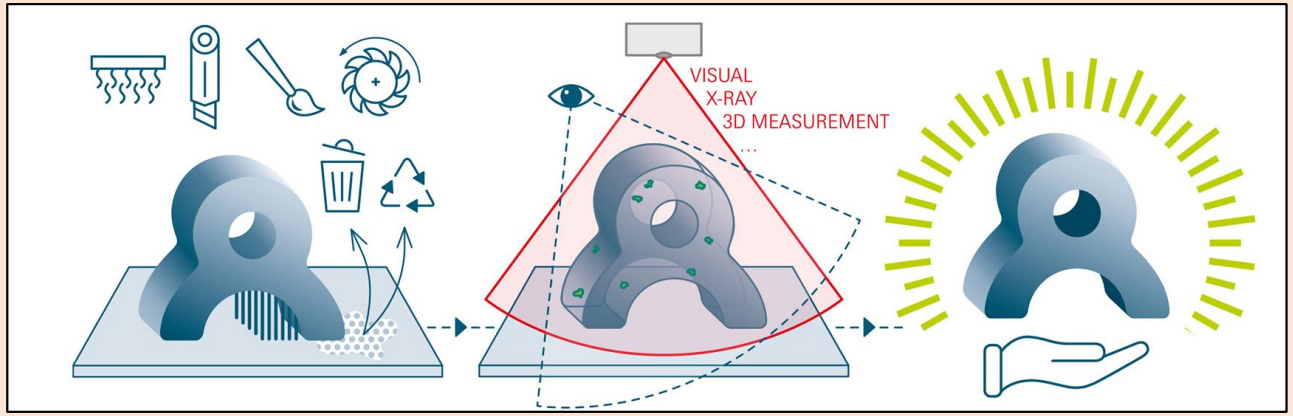
CAE (Computer-Aided Engineering) – data harvesting - structural stresses, deformations, displacement, and failures – **FEA - CFD**





CAM

Computer-Aided Manufacturing



Pós Processamento

Testes, Medições & controle de qualidade

https://formnext.mesago.com/frankfurt/en/industry-insights/am-field-guide/process_in_general.html

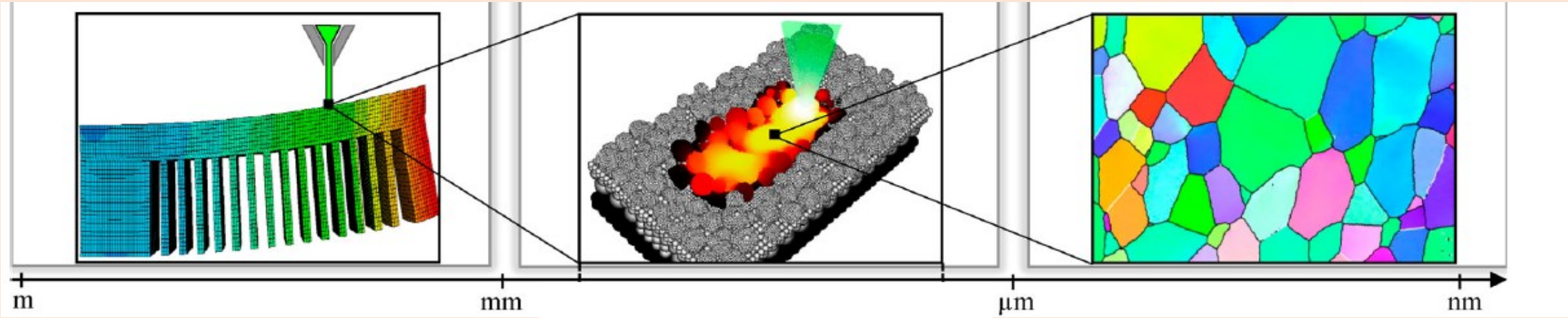
Escalas de modelagem

O que modelar?

Macro-scale models
heat flow and structural (stress) deformation & interlayer bonding,

Meso-scale models
pool geometry, bead formation, and dilution.

Micro-scale phase-field
grain growth, microstructural transitions, solidification kinetics



Distorção, stress residual

Métodos: **FEM, FVM**

Fluxo, poça de fusão

Métodos: **CFD, SPH, LBM**

Grãos, nucleação, vazios

Métodos: **CA, MC, PF**

https://formnext.mesago.com/frankfurt/en/industry-insights/am-field-guide/process_in_general.html

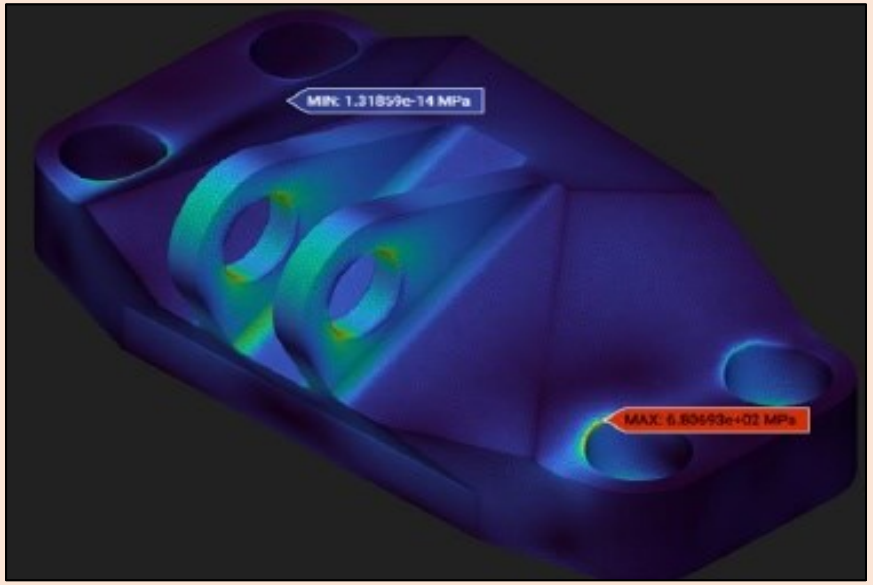
FEA – Análise de Elementos Finitos

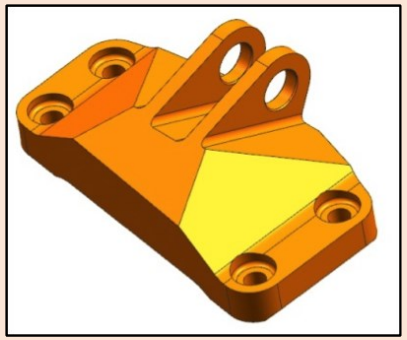
Como modelar?

- Decompõe um objeto 3D complexo em milhares de pequenas partes gerenciáveis ("elementos"); - Calcula tensões estruturais, falhas e deformações.

Stress, Deformação, Vibração, Fadiga

Análise Estrutural

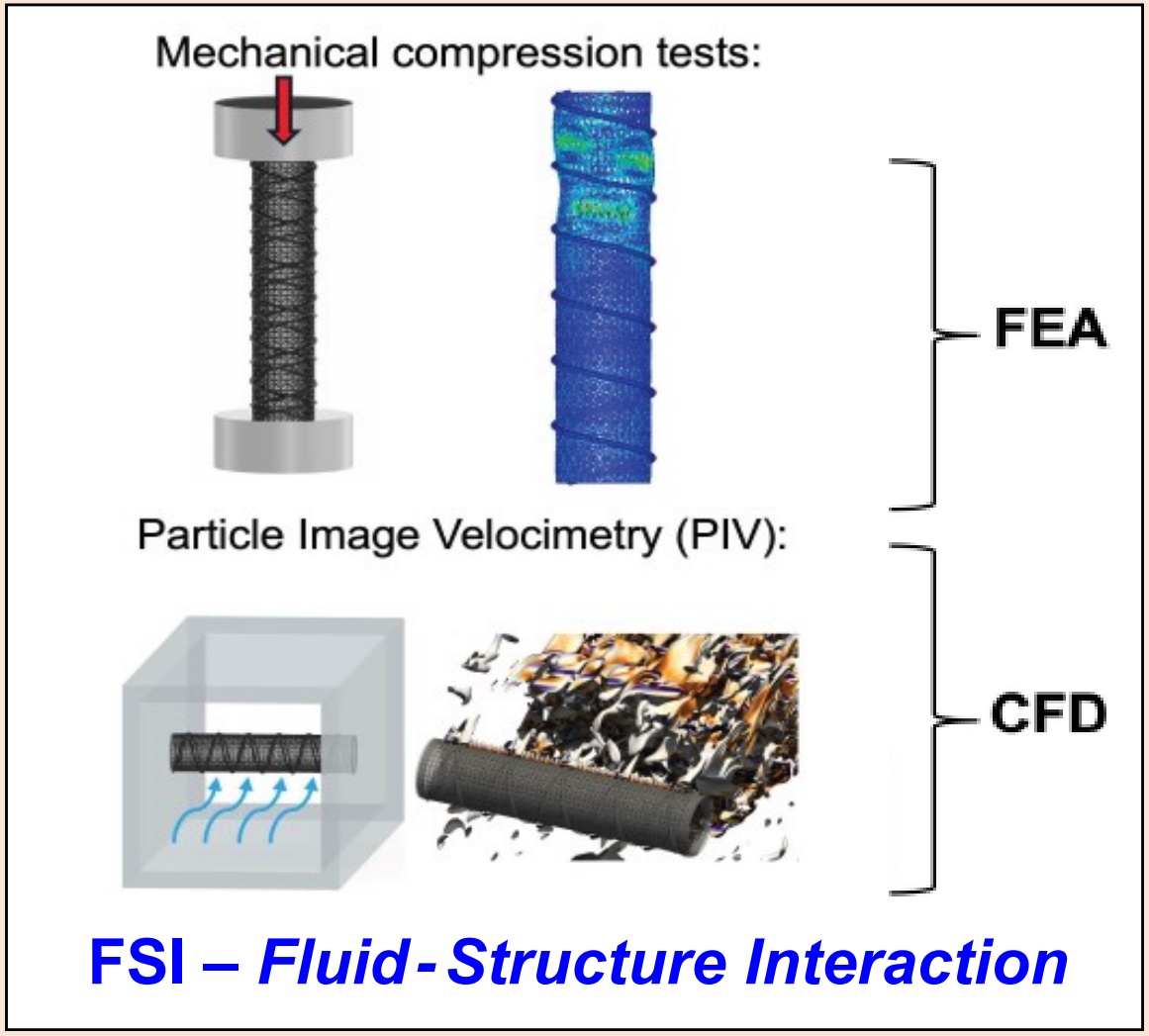




Peça de Suporte

Fluxo, pressão, temperatura, momento, energia

CFD - Fluidodinâmica Computacional

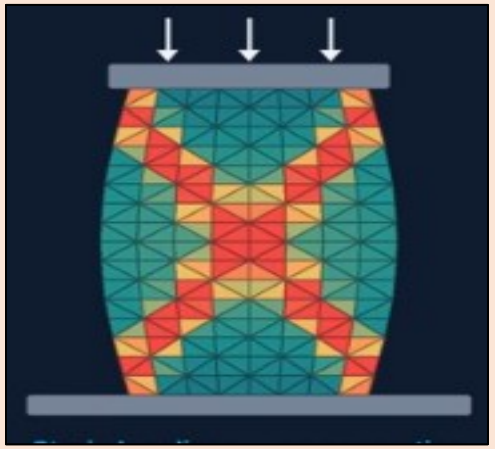


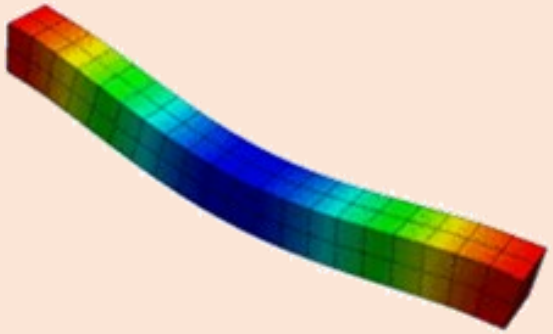
Como modelar?

FEM Finite Element Method

Material contínuo

stress e deformação

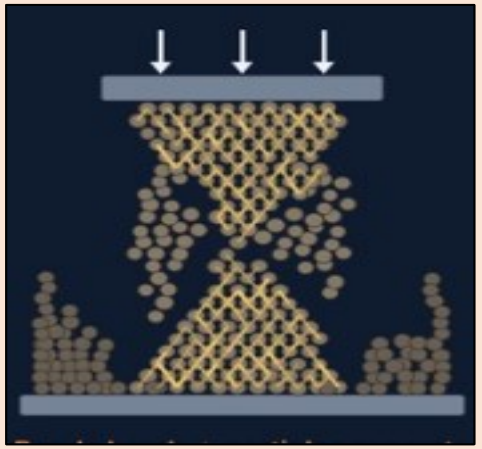


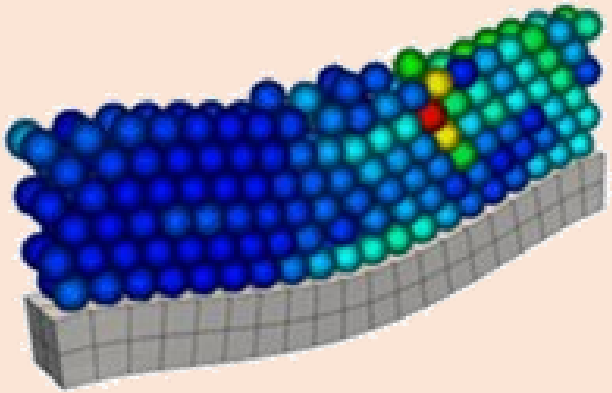


DEM Discrete Element Method

Interação de partículas

Escorregamento, aberturas

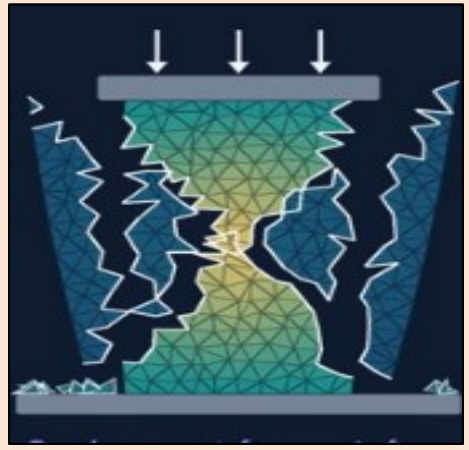




FDEM Finite-Discrete Element Method

Combinação: = fraturas

deformação + interação de blocos



FDM

Finite-difference

MODFLOW

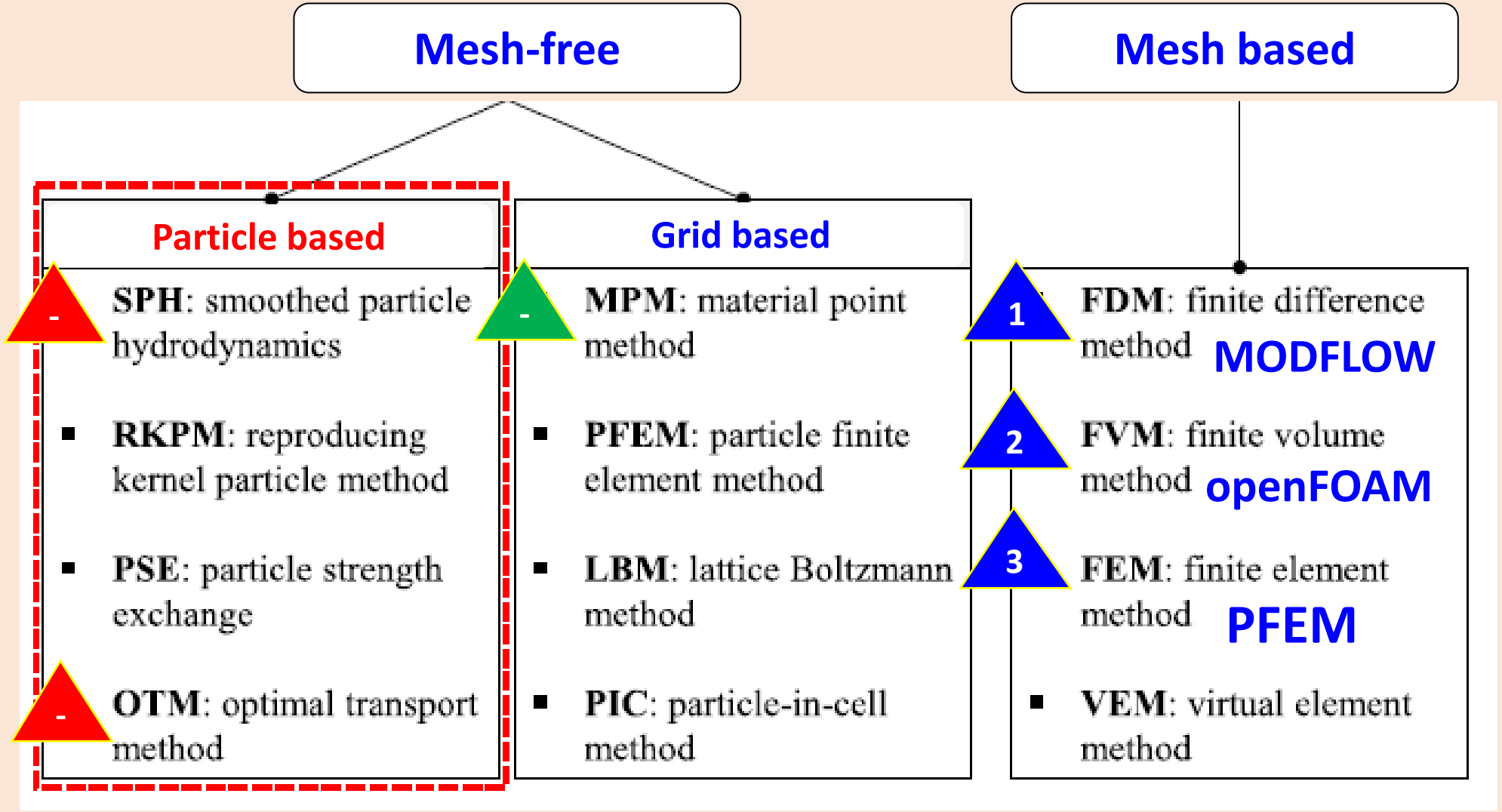
FVM

Finite-volume

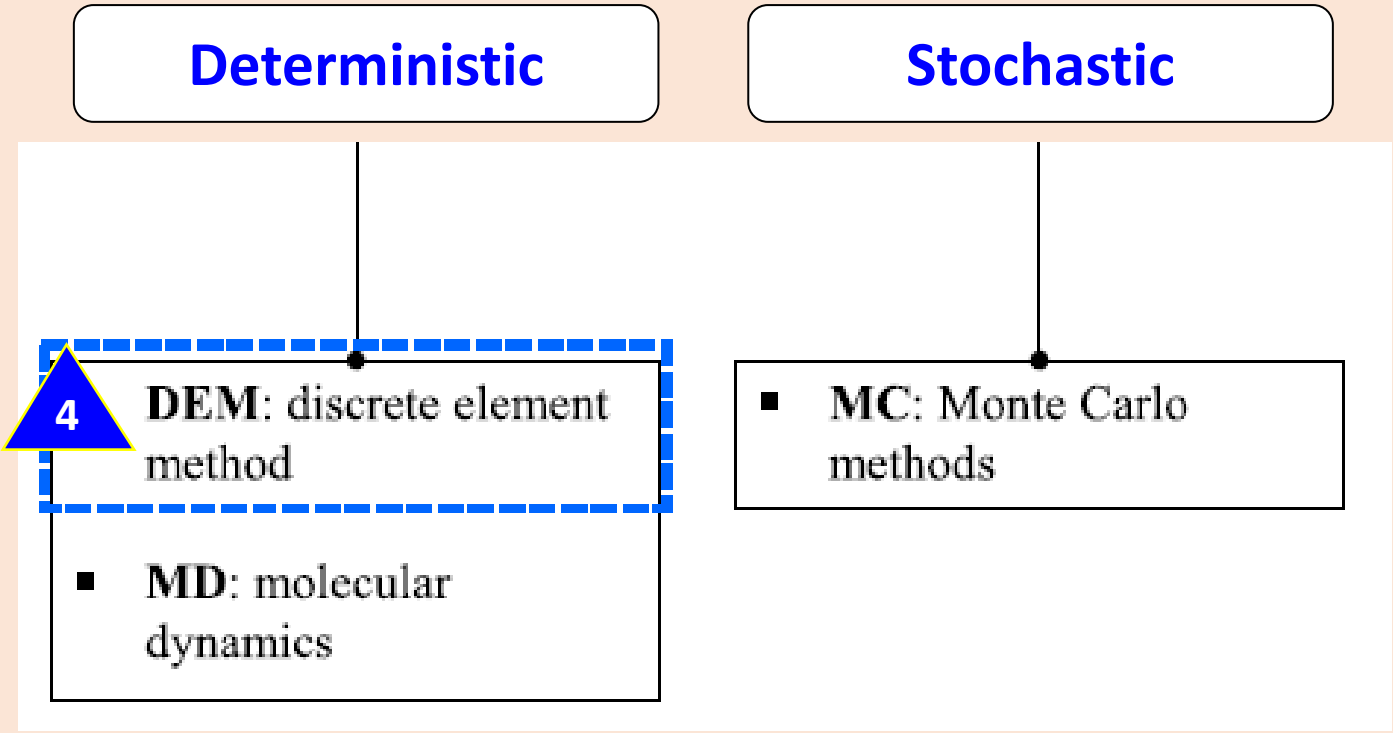
OpenFOAM, FiPy

Como modelar?

Massive parallelisation



Como modelar?



ML algorithms

physics-based **surrogate models**

01 - finite element analysis **(FEA)**

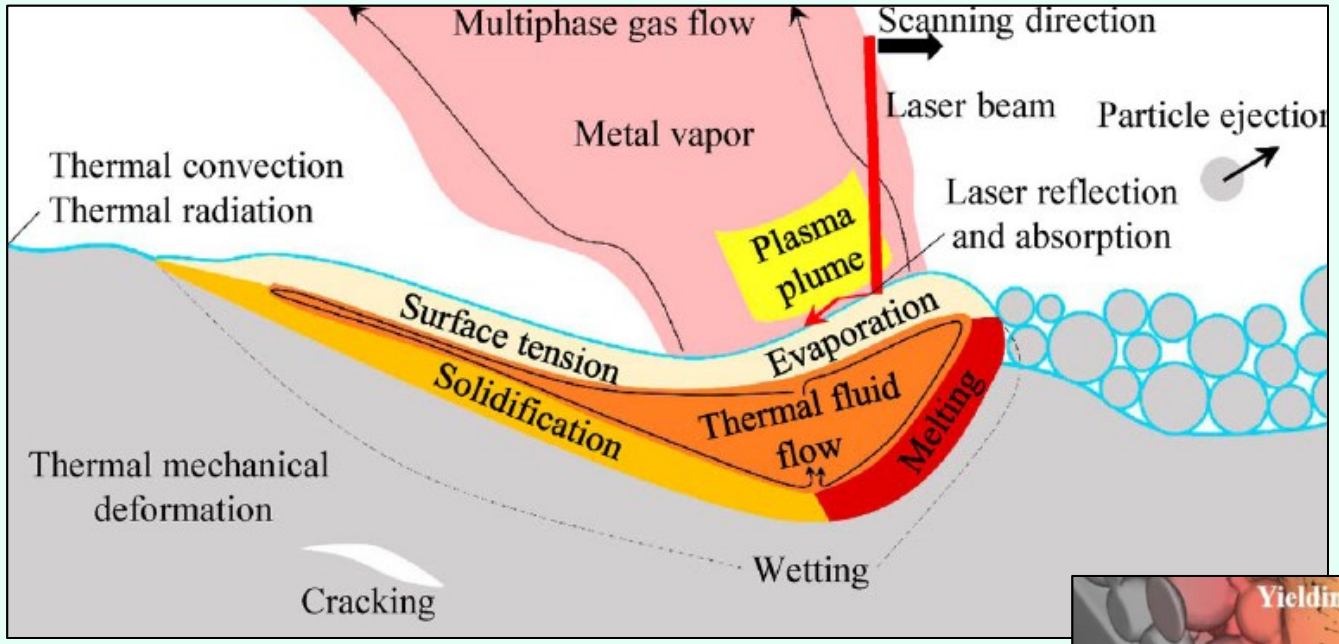
02 - **phase-field modeling** techniques

Macro

Modelling and simulation of metal additive manufacturing processes **with particle methods.**

PDF

- simulation-informed
- closed-loop control feedback
- real-time monitoring systems



Meso-scale

Mult—physics
Mult—dimensional

$$\nabla . \mathbf{V} = 0$$

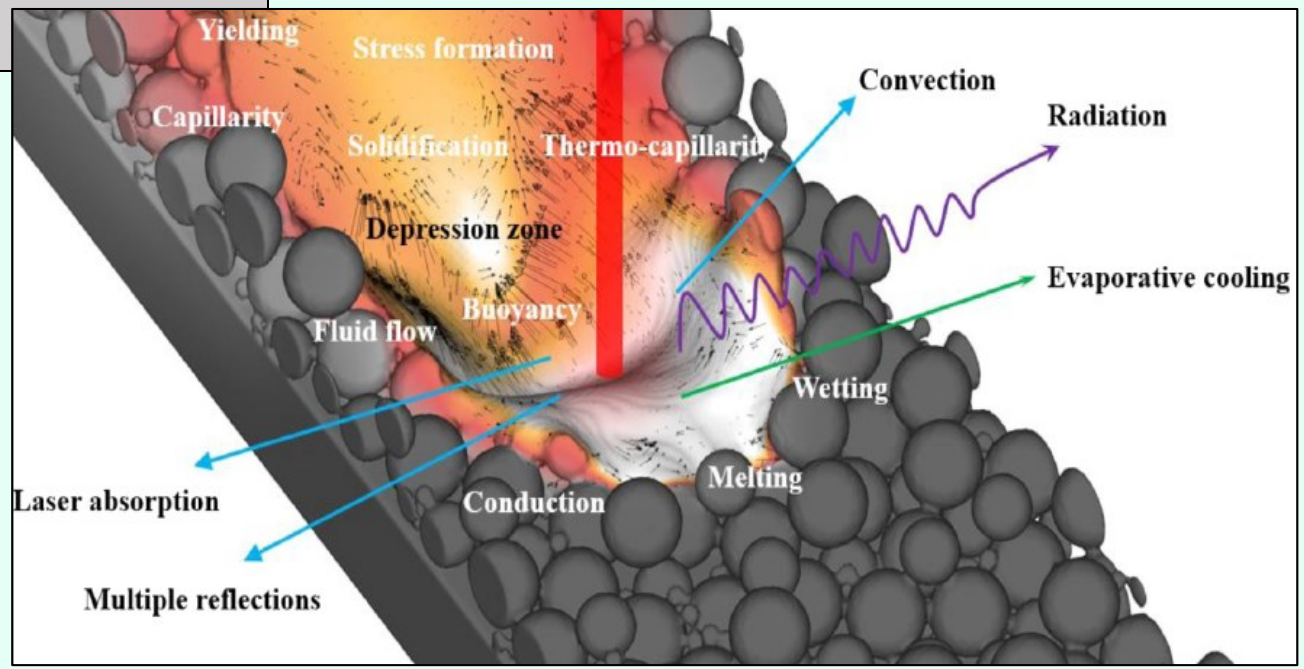
Continuidade

$$\rho \frac{\partial \mathbf{V}}{\partial t} = \rho \mathbf{g} - \nabla \mathbf{p} + \mu \nabla^2 \mathbf{V}$$

Qde de movimento

$$\rho C_V \frac{\partial T}{\partial t} = K \nabla^2 T + \Phi$$

Energia



CFD Methodology

https://github.com/DLR-RY/VoFLibrary

CFD

Incompressible, multiphase flow with surface tension and phase change

Volume of fluid method (VOF)

$$\frac{\partial \alpha}{\partial t} + \nabla . (\alpha u) = 0$$

Volume of fraction

$\alpha = V_i / V$

α = 0 (solid)
α = 1 (liquid)
$0 < \alpha < 0$ (interface)

Continuity Equation (mass conservation)

$$\nabla . u = 0$$

Surface tension Marangoni force

Momentum conservation

$$\underbrace{\frac{\partial(\alpha \mathbf{u})}{\partial t}}_{\text{Inertia (rate of change of momentum)}} + \nabla \cdot (\alpha\, \mathbf{u} \otimes \mathbf{u}) = \underbrace{-\nabla p}_{\text{Pressure force}} + \underbrace{\nabla \cdot \tau}_{\text{Viscous force}} + \underbrace{\rho \mathbf{g}}_{\text{Gravity / Body force}} + \underbrace{\sigma \kappa \nabla \alpha + \nabla \sigma \cdot (\mathbf{I} - \mathbf{nn})}_{\text{Surface tension + Marangoni force}}$$

Laser heat input **latent input**

Energy conservation

$$\underbrace{\rho c_p \frac{\partial T}{\partial t}}_{\text{Local (transient) change in temperature}} + \underbrace{\rho c_p \mathbf{u} \cdot \nabla T}_{\text{Convective heat transport}} = \underbrace{\nabla \cdot (k \nabla T)}_{\text{Heat conduction}} + \underbrace{Q_{\text{laser}}}_{\text{Laser heat input}} + \underbrace{Q_{\text{phase}}}_{\text{Latent heat (phase change)}}$$

"… the basis of how the simulation solver is modelled and how we can rely on the data that has been predicted … based on this robust physic equations that are provided "

https://www.youtube.com/live/J4ieku-yXrY

EX: arc welding – analytical solution

Directed Energy Deposition (DED - ARC)

Table 1 Parameters of the welding process

Voltage (V)	Welding current (A)	Speed (mm/s)	Shielding gas	Flow rate (cfh)	Electrode diameter (mm)	Arc length (mm)
14.6	150	1.5	Argon	40	2.4	3

$$\rho c\left(\frac{\partial T}{\partial t}+v\,\nabla T\right)=\nabla.(\mathrm{k}\nabla\mathrm{T})+\dot{Q}$$

Convection Diffusion

new (x) coordinate system

$$\frac{\rho c v}{\mathrm{k}}\frac{\partial \mathrm{T}}{\partial \xi}=\nabla^2 T+\frac{\dot{Q}}{k}$$

$$\xi = y - vt$$

$$T = T_0 + \frac{Q}{2\pi kR} e^{\left(-\frac{\rho c v}{2k}(R+\xi)\right)}$$

$$R=\sqrt{x^2+\xi^2+z^2}$$

Excel

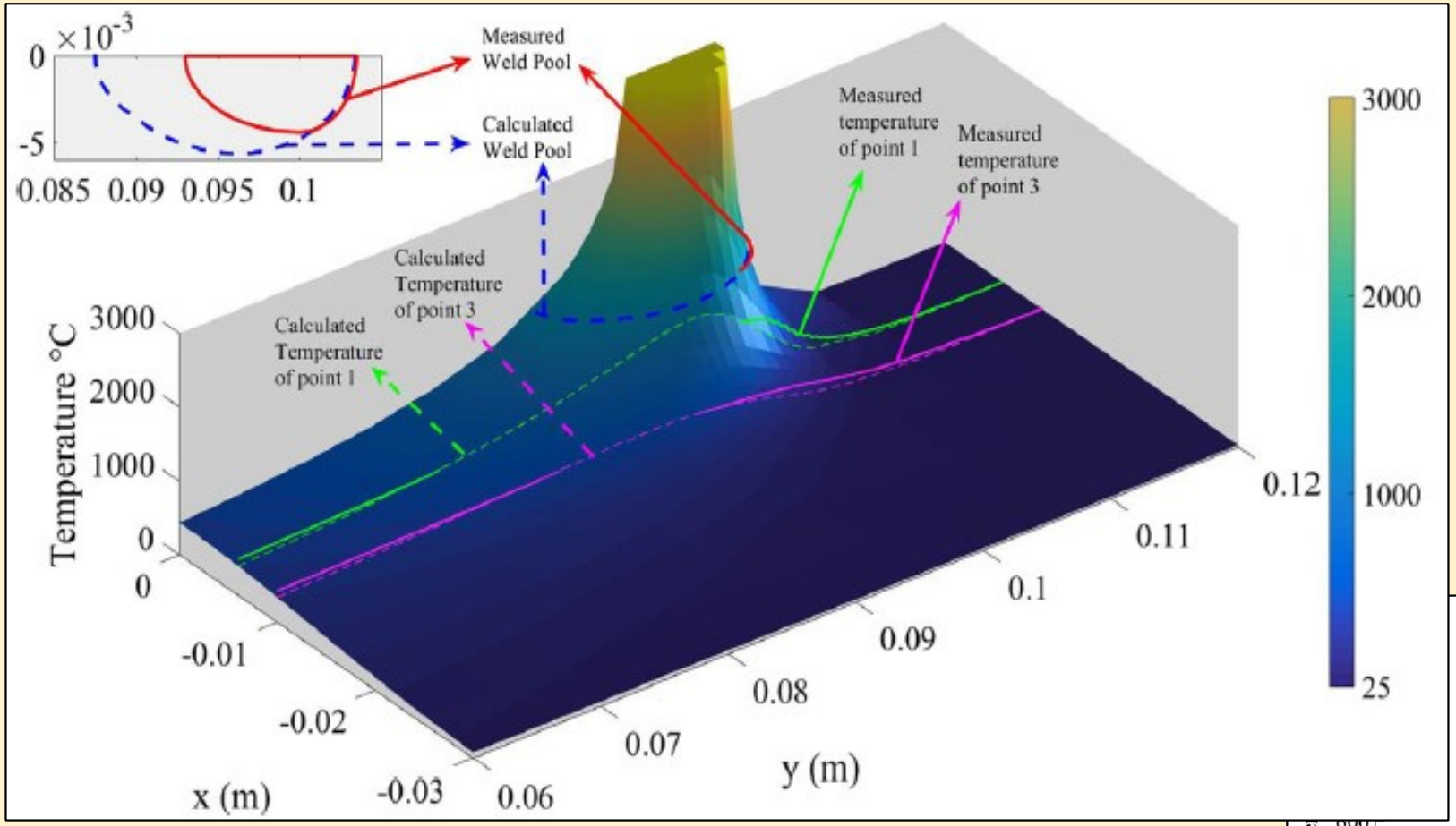


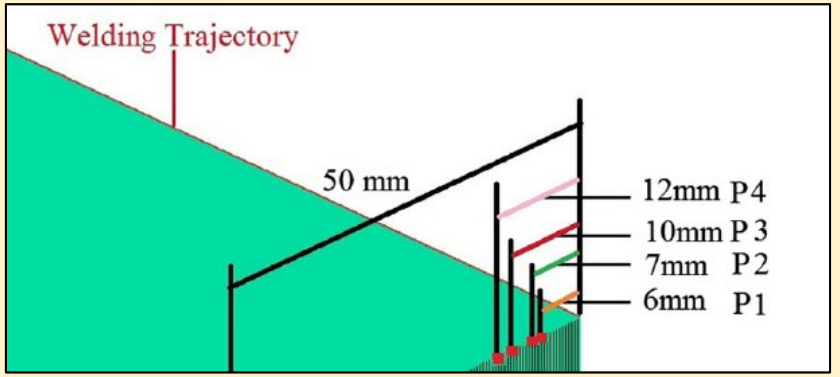
- k Heat conductivity
- R Distance - moving heat source and pt interest
- c Specific heat
- v Welding speed
- ξ Moving coordinate parallel to the welding

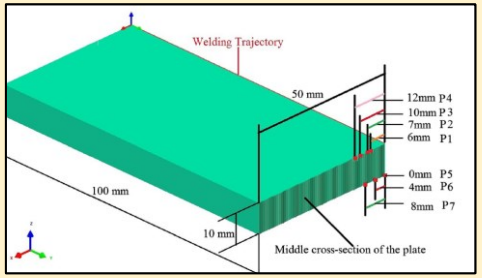
An analytical solution for temperature distribution in fillet arc welding based on an adaptive function Bagher & Norbert Enzinger

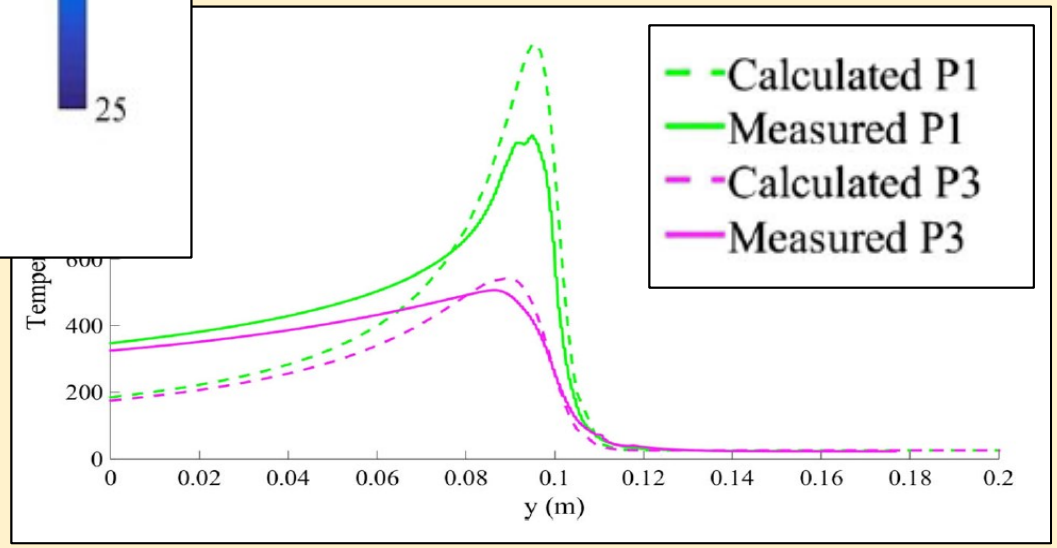
EX: arc welding - analytical solution

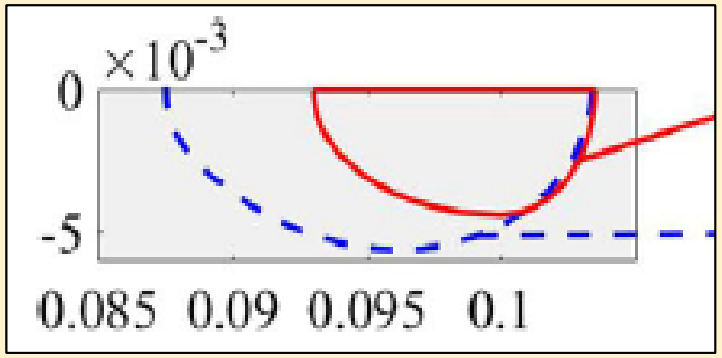
Directed Energy Deposition (DED - ARC)











$$T = T_0 + \frac{Q}{2\pi kR} e^{\left(-\frac{\rho c v}{2k}(R+\xi)\right)}$$

Part 01

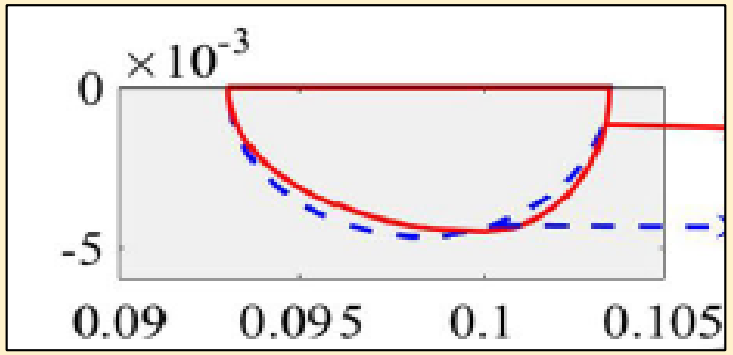
EX:

$$T = T_0 + \frac{Q}{2\pi kR} e^{\left(-\frac{\rho c v}{2k}(R+\xi)\right)}$$

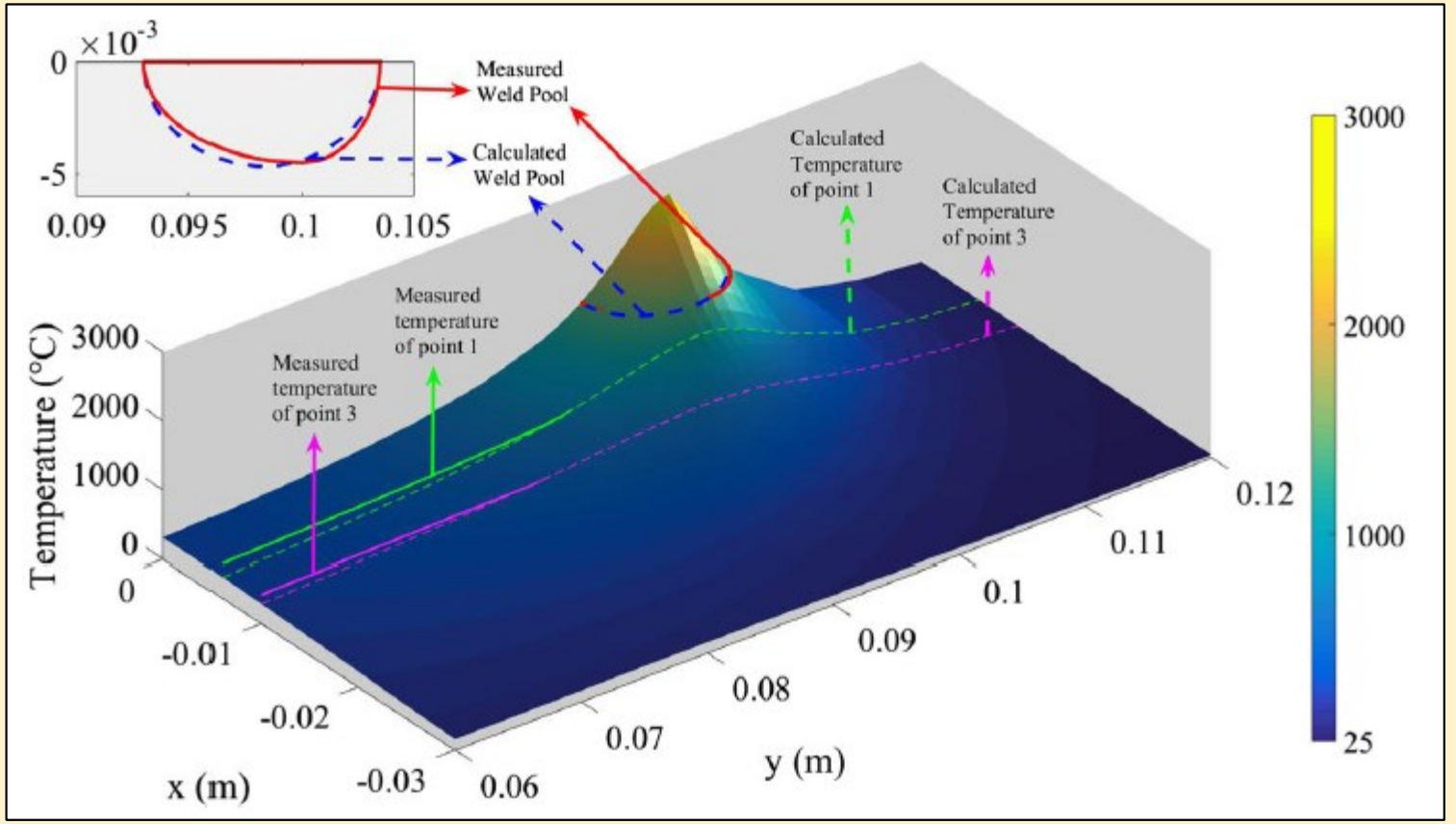
$$R_p^* = \sqrt{\left(\frac{x}{a}\right)^2 + \left(\frac{\xi}{b}\right)^2 + \left(\frac{z}{c}\right)^2 + d^2}$$

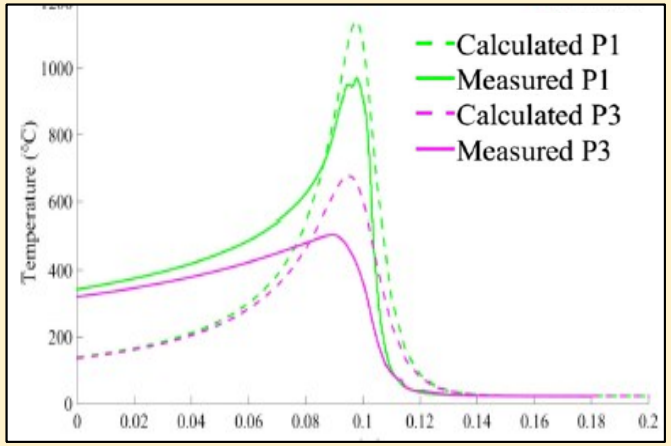
$$T = T_0 + \frac{1}{\sqrt{\left(\frac{x}{a_m}\right)^2 + \left(\frac{\xi}{b_m}\right)^2 + \left(\frac{z}{c_m}\right)^2 + {d_m}^2}} e^{(-B(R+\xi))}$$

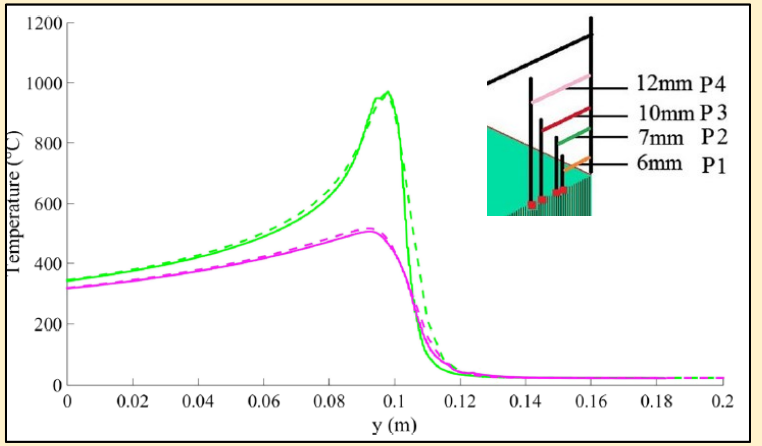
$$a_m = \frac{Q}{2\pi k} \times a; b_m = \frac{Q}{2\pi k} \times b; c_m = \frac{Q}{2\pi k} \times c; d_m$$



Part 02







EX: Laser Powder Bed Fusion (L PBF)

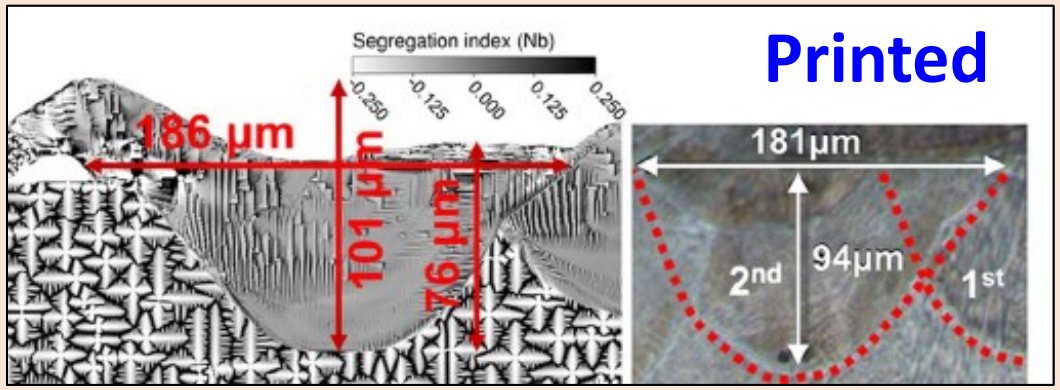
nature communications

Article

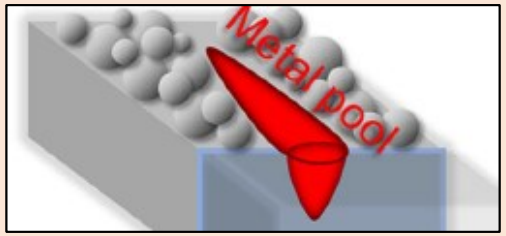
https://doi.org/10.1038/s41467-023-43563-x

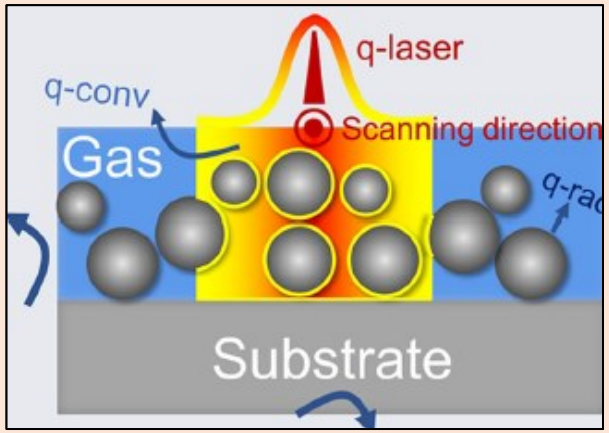
Micro-scale

Solute trapping and non-equilibrium microstructure during rapid solidification of additive manufacturing

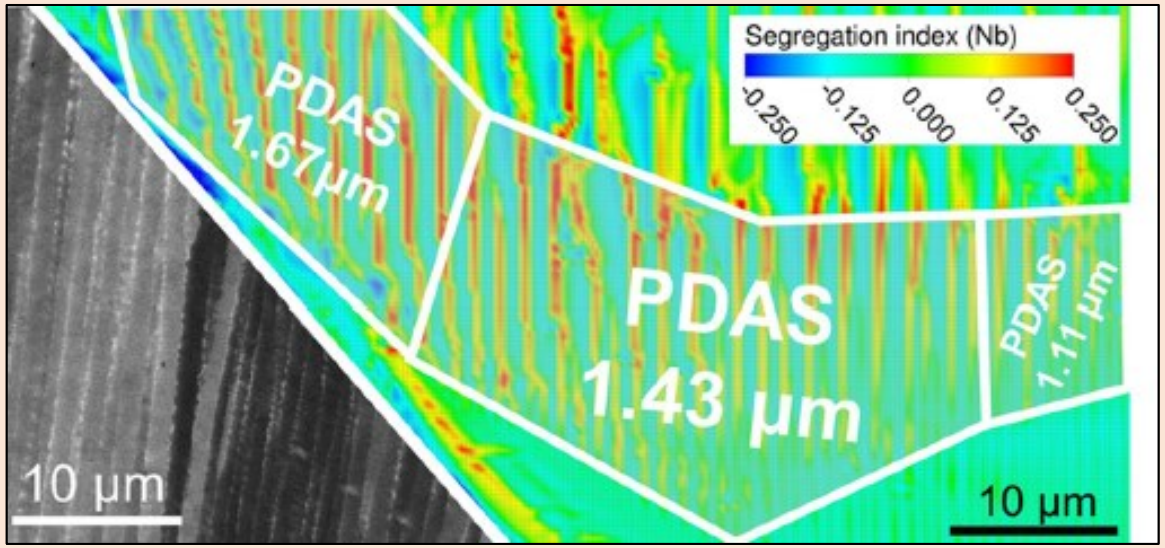


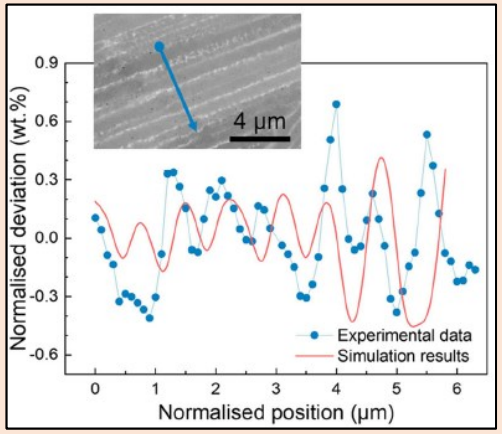
- insight into the solute transport process;
- high-fidelity physics-based numerical modelling;
- effect of melt convection and solute partition.





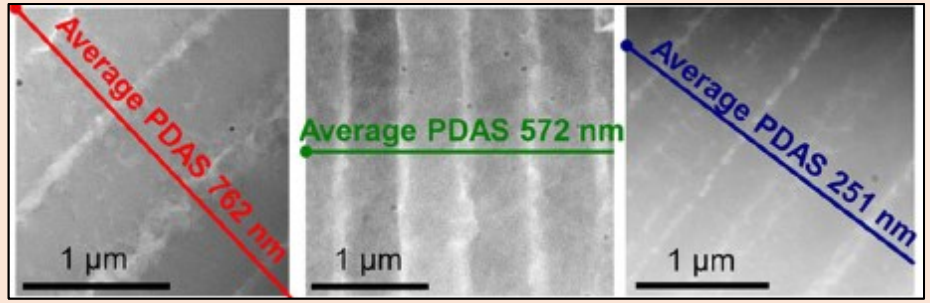
Model





Temperature, flow velocity, solute concentration, and microstructure

... directly coupled



210 × 400 µm computational domain (2D) 1,341,561 grid 0.25 µm

Obrigado!